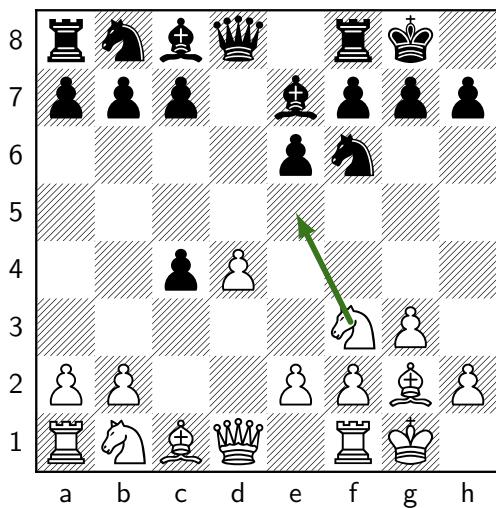


# Melanie-BASE — Catalan

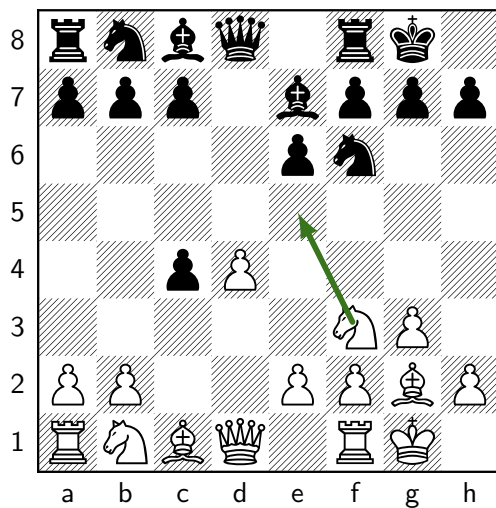
Melanie (Deep\_Vision)

## Abstract

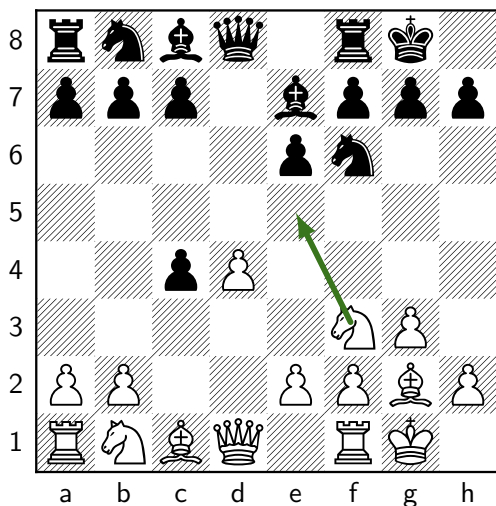
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



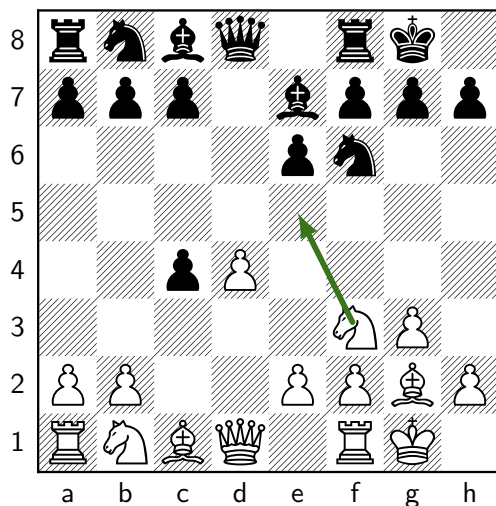
(a)



(b)

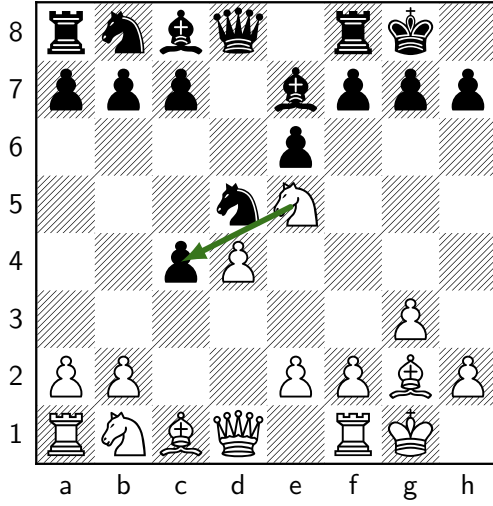


(c)

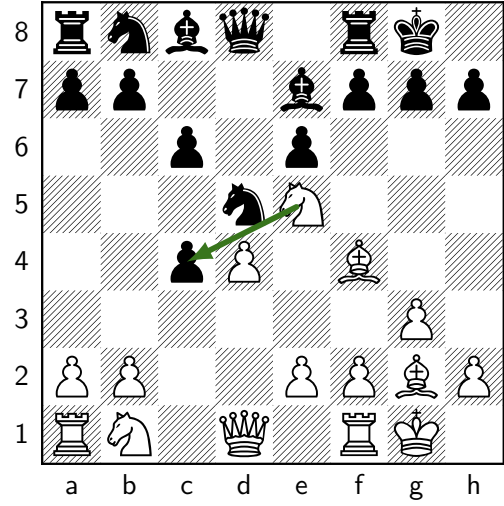


(d)

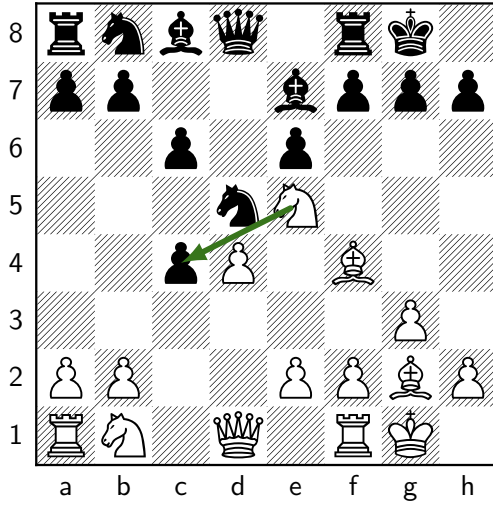
Figure 1: **Move 7, White: Ne5**. (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3464` (ply 12) [structural distance  $d=0$ , QAP obj=74]: the master played Ne5; under  $\mu$  our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3465` (ply 12) [structural distance  $d=0$ , QAP obj=74]: the master played Ne5; under  $\mu$  our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3065` (ply 12) [structural distance  $d=0$ , QAP obj=74]: the master played Ne5; under  $\mu$  our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ne5 (summed QAP objective 928, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



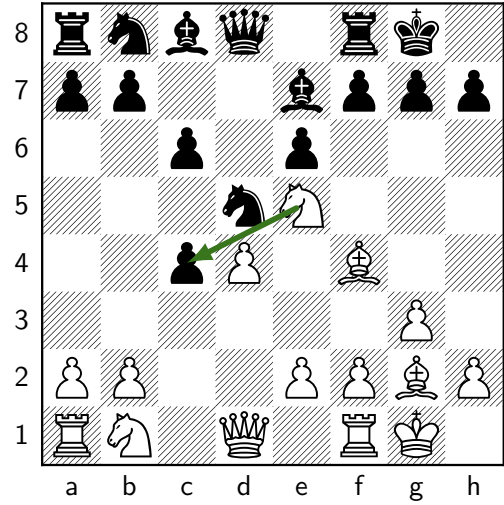
(a)



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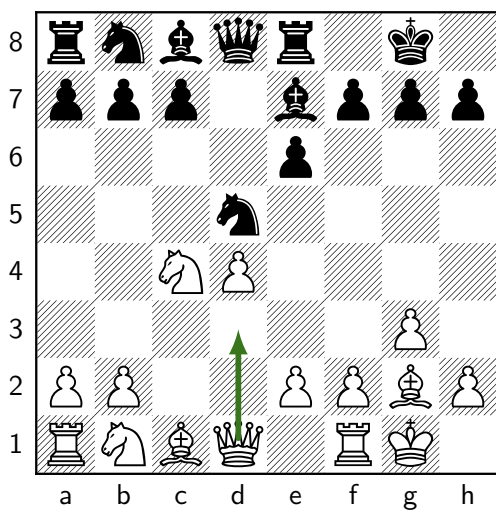


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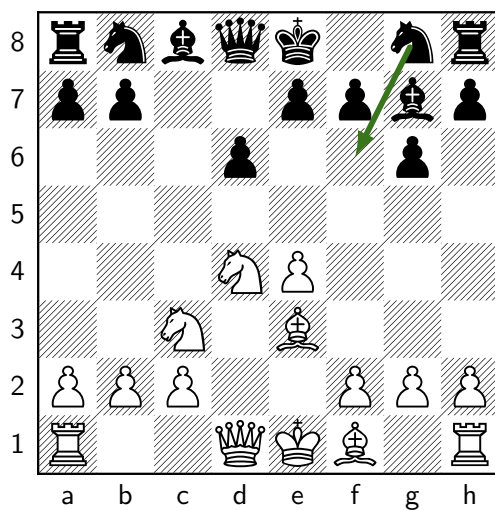


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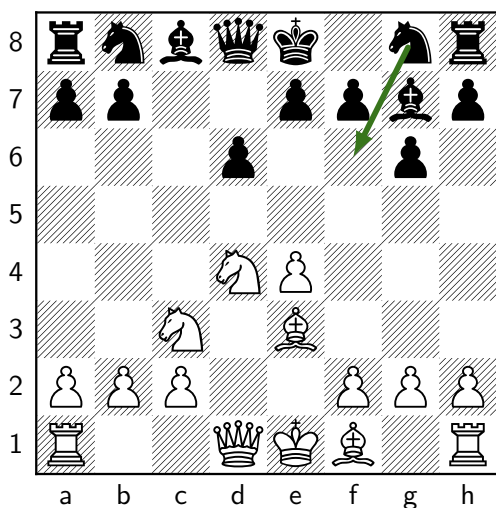
Figure 2: **Move 8, White: Nxc4**. (a) Our position; Melanie plays Nxc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#271` (ply 18) [structural distance  $d=33$ , QAP obj=43]: the master played Nxc4; under  $\mu$  our Ne5 plays the role of that game’s Ne5; cross-type roles: our be7 as their Pg3, our Bc1 as their Pd4; transferred move e5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#272` (ply 18) [structural distance  $d=33$ , QAP obj=43]: the master played Nxc4; under  $\mu$  our Ne5 plays the role of that game’s Ne5; cross-type roles: our be7 as their Pg3, our Bc1 as their Pd4; transferred move e5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3736` (ply 18) [structural distance  $d=33$ , QAP obj=43]: the master played Nxc4; under  $\mu$  our Ne5 plays the role of that game’s Ne5; cross-type roles: our be7 as their Pg3, our Bc1 as their Pd4; transferred move e5c4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nxc4 (summed QAP objective 261, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



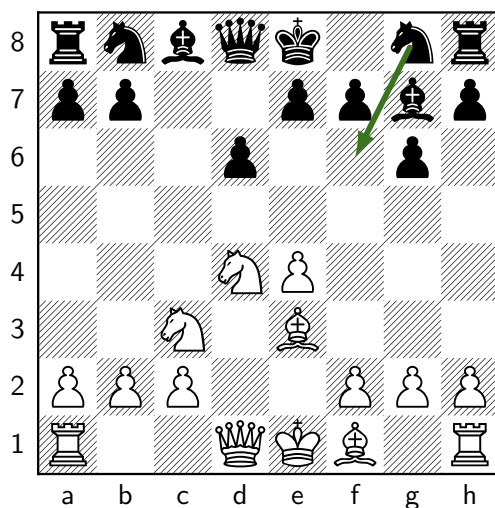
(a)



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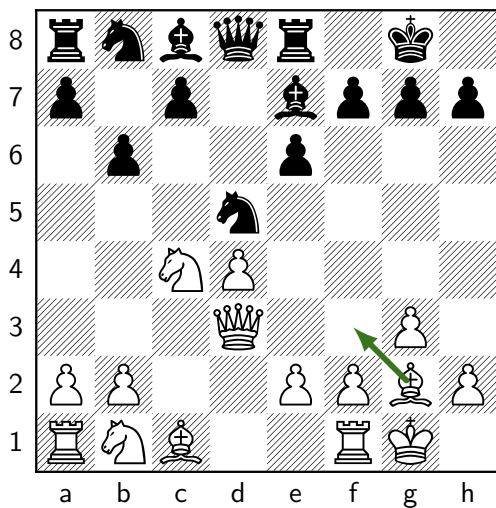


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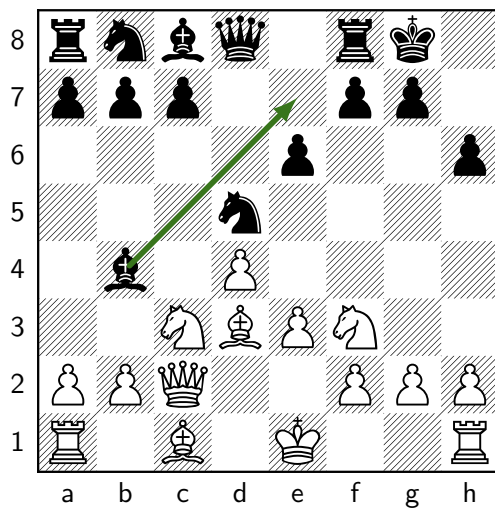


(d)

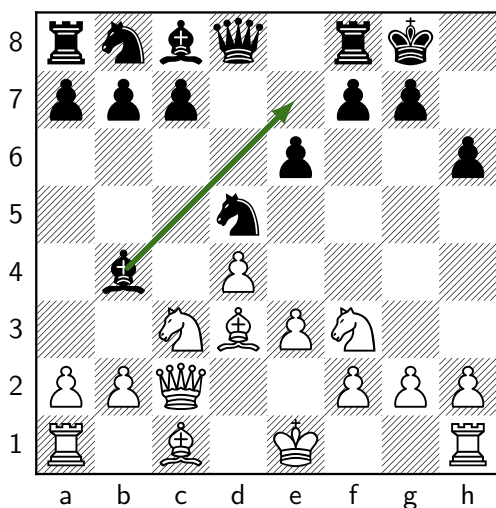
Figure 3: **Move 9, White: Qd3.** (a) Our position; Melanie plays Qd3 (d1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1092` (ply 11) [structural distance  $d=50$ , QAP obj=24]: the master played Nf6; under  $\mu$  our Qd1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra8, our Pf2 as their Nb8; transferred move d1d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1093` (ply 11) [structural distance  $d=50$ , QAP obj=24]: the master played Nf6; under  $\mu$  our Qd1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra8, our Pf2 as their Nb8; transferred move d1d3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1116` (ply 11) [structural distance  $d=50$ , QAP obj=24]: the master played Nf6; under  $\mu$  our Qd1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra8, our Pf2 as their Nb8; transferred move d1d3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Qd3 (summed QAP objective 317, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



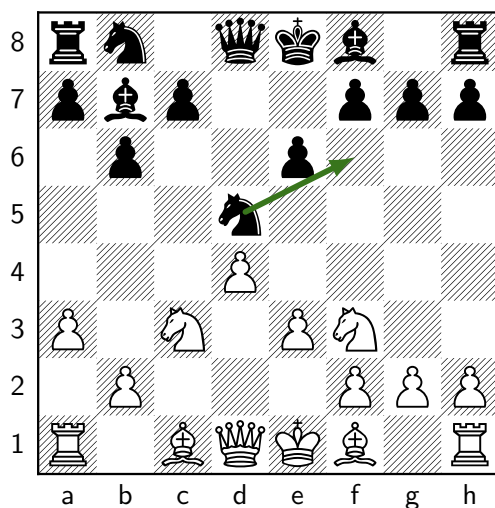
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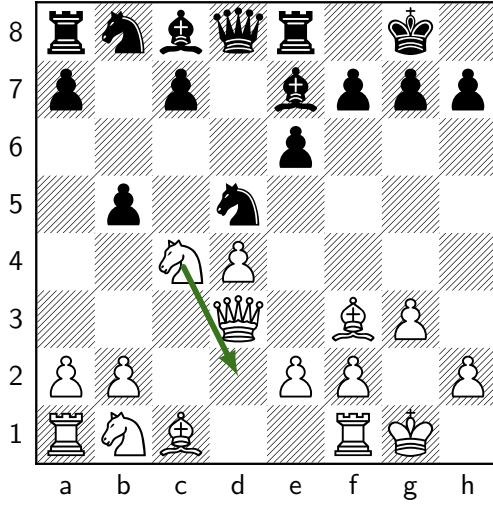


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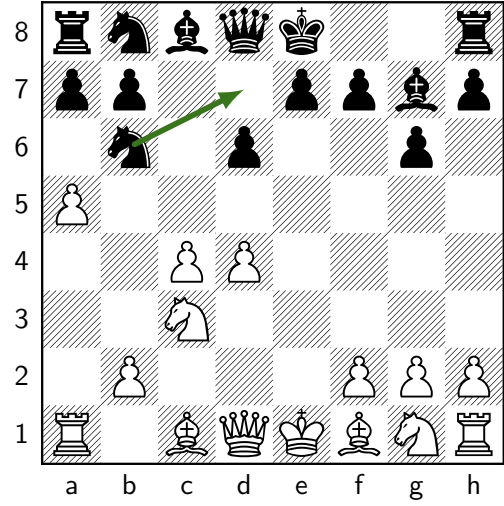


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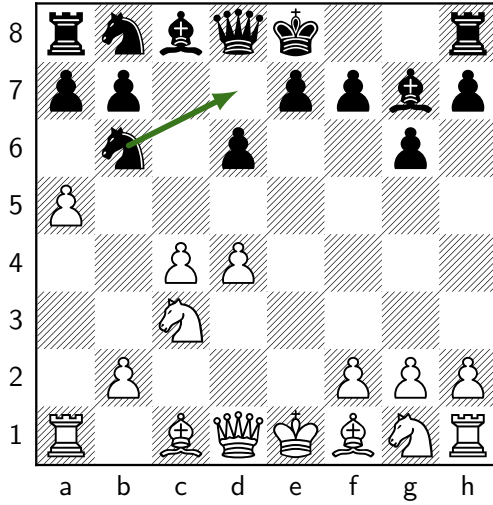
Figure 4: **Move 10, White: Bf3**. (a) Our position; Melanie plays Bf3 (g2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2378` (ply 15) [structural distance  $d=62$ , QAP obj=23]: the master played Be7; under  $\mu$  our Bg2 plays the role of that game’s Bb5; cross-type roles: our Nc4 as their Ra8, our Ra1 as their Nb8; transferred move g2f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2379` (ply 15) [structural distance  $d=62$ , QAP obj=23]: the master played Be7; under  $\mu$  our Bg2 plays the role of that game’s Bb5; cross-type roles: our Nc4 as their Ra8, our Ra1 as their Nb8; transferred move g2f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1883` (ply 13) [structural distance  $d=59$ , QAP obj=20]: the master played Nf6; under  $\mu$  our Bg2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our pe6 as their Ra8, our pb6 as their Nb8; transferred move g2f3 (**endorses this move**). Synthesis: 27 retrieved analogs transferred to Bf3 (summed QAP objective 554, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



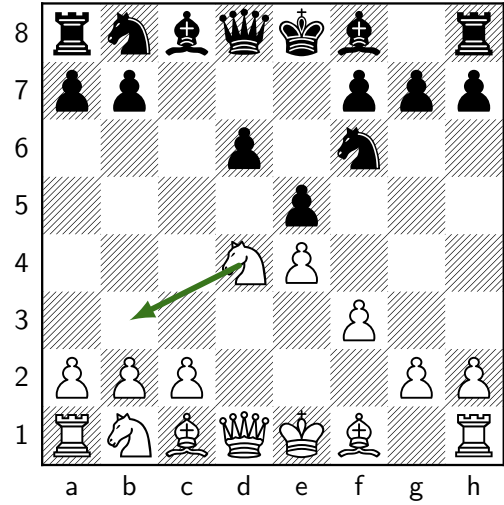
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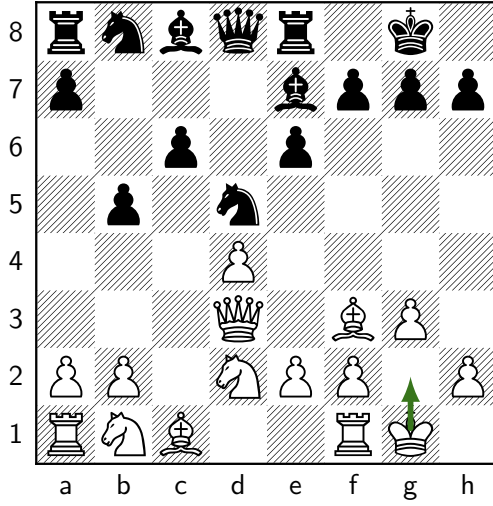


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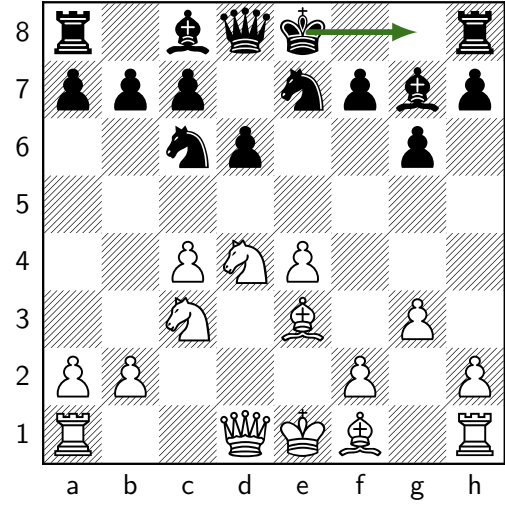


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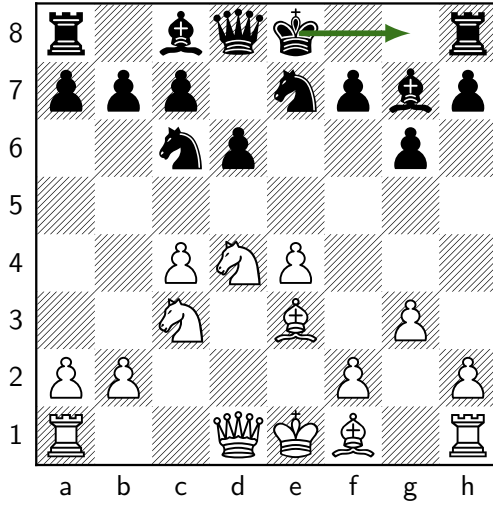
Figure 5: **Move 11, White: Ncd2.** (a) Our position; Melanie plays Ncd2 (c4d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#505` (ply 15) [structural distance  $d=62$ , QAP obj=28]: the master played N6d7; under  $\mu$  our Nc4 plays the role of that game’s Nb3; cross-type roles: our Pb2 as their Ra8, our pf7 as their Nb8; transferred move c4d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#506` (ply 15) [structural distance  $d=62$ , QAP obj=28]: the master played N6d7; under  $\mu$  our Nc4 plays the role of that game’s Nb3; cross-type roles: our Pb2 as their Ra8, our pf7 as their Nb8; transferred move c4d2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#604` (ply 10) [structural distance  $d=63$ , QAP obj=24]: the master played Nb3; under  $\mu$  our Nc4 plays the role of that game’s Nd4; cross-type roles: our Qd3 as their Nb1, our pe6 as their Bc1; transferred move c4d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ncd2 (summed QAP objective 105, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



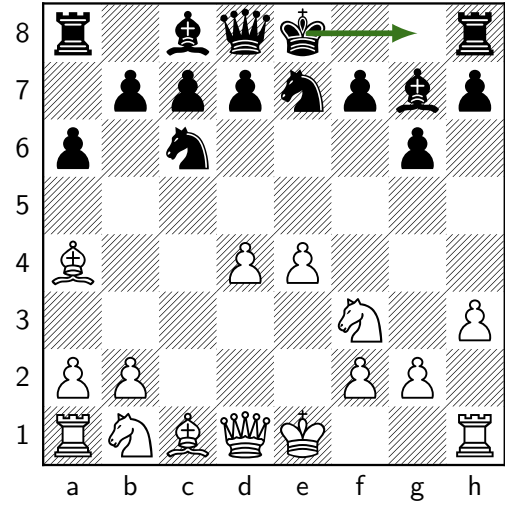
(a)



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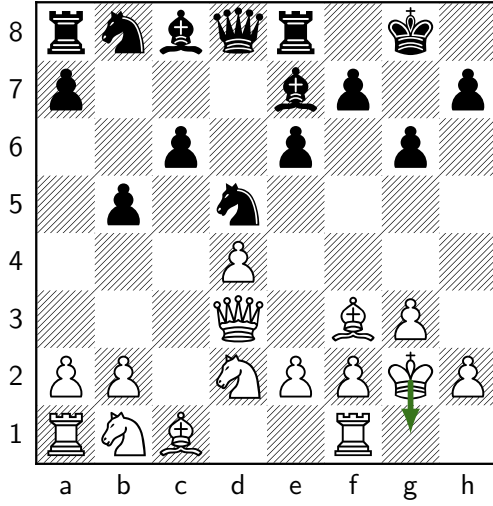


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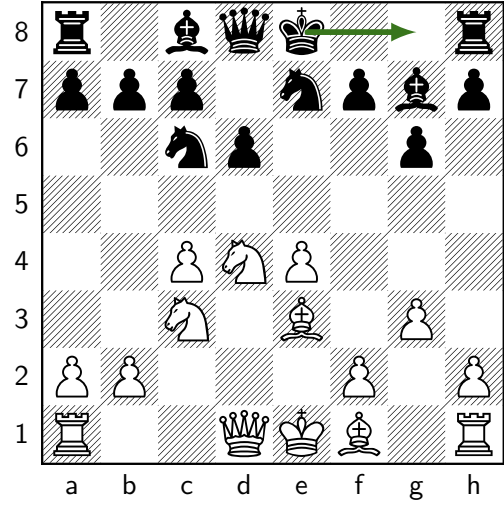


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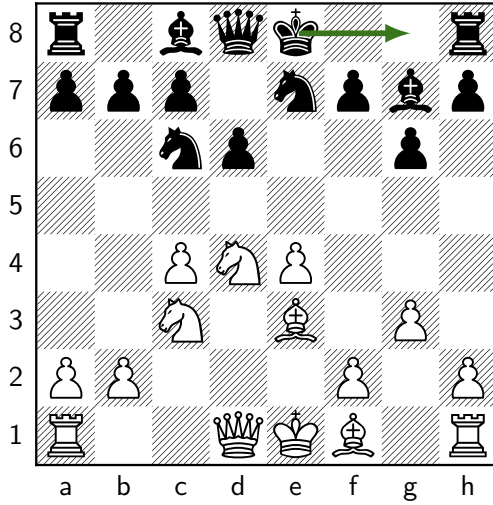
Figure 6: **Move 12, White: Kg2.** (a) Our position; Melanie plays Kg2 (g1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1597` (ply 15) [structural distance  $d=76$ , QAP obj=33]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pf2* as their *Ra8*, our *Qd3* as their *Bc8*; transferred move g1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1598` (ply 15) [structural distance  $d=76$ , QAP obj=33]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pf2* as their *Ra8*, our *Qd3* as their *Bc8*; transferred move g1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3791` (ply 15) [structural distance  $d=85$ , QAP obj=25]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb1* as their *Ra8*, our *Pd4* as their *Bc8*; transferred move g1g2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kg2 (summed QAP objective 191, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



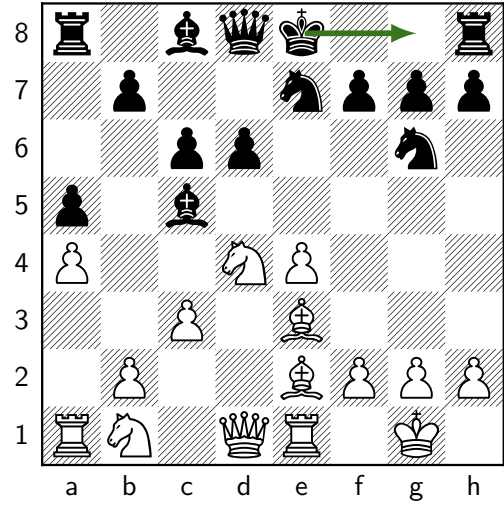
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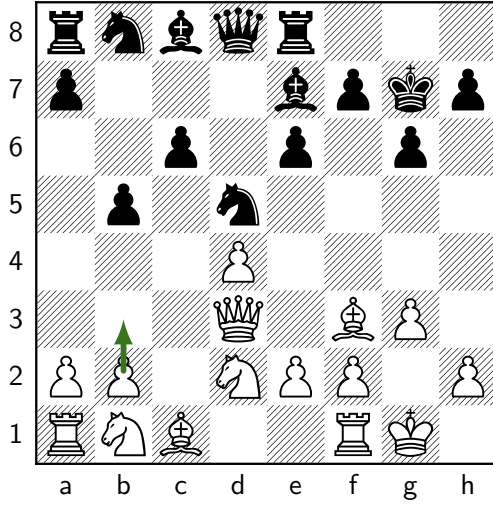
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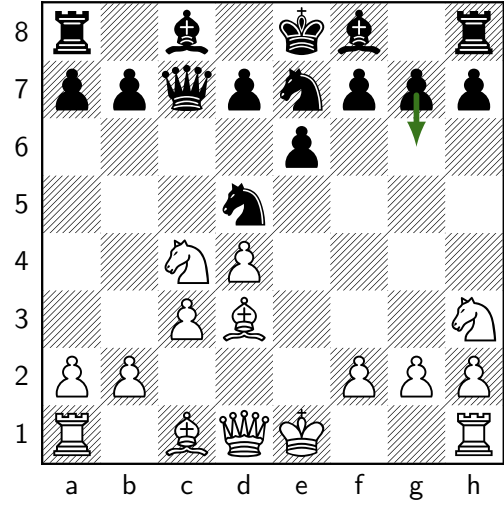
(d)

Figure 7: **Move 13, White: Kg1 (g2g1).** (a) Our position; Melanie plays Kg1 (g2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1597` (ply 15) [structural distance  $d=97$ , QAP obj=32]: the master played 0-0; under  $\mu$  our Kg2 plays the role of that game's Ke1; cross-type roles: our bc8 as their Ra8, our Pe2 as their Bc8; transferred move g2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1598` (ply 15) [structural distance  $d=97$ , QAP obj=32]: the master played 0-0; under  $\mu$  our Kg2 plays the role of that game's Ke1; cross-type roles: our bc8 as their Ra8, our Pe2 as their Bc8; transferred move g2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2773` (ply 23) [structural distance  $d=89$ , QAP obj=13]: the master played 0-0; under  $\mu$  our Kg2 plays the role of that game's Ke1; cross-type roles: our be7 as their Ra8, our pe6 as their Bc8; transferred move g2g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

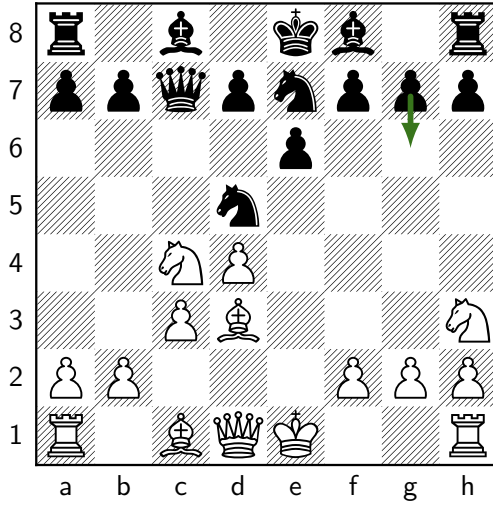




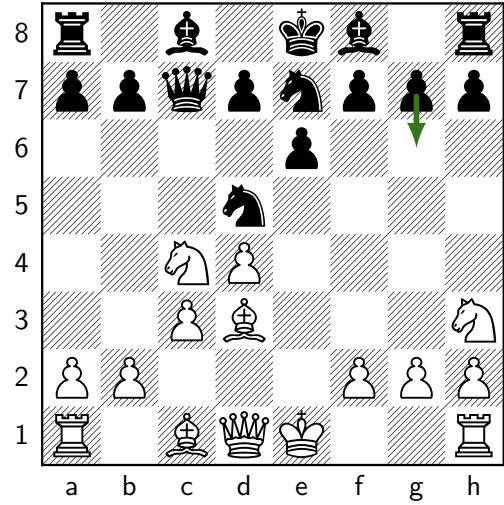
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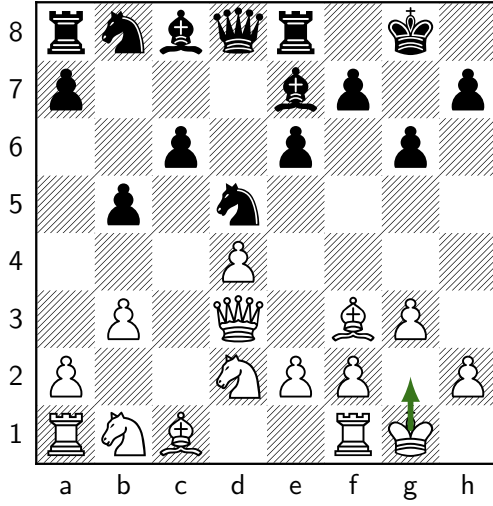


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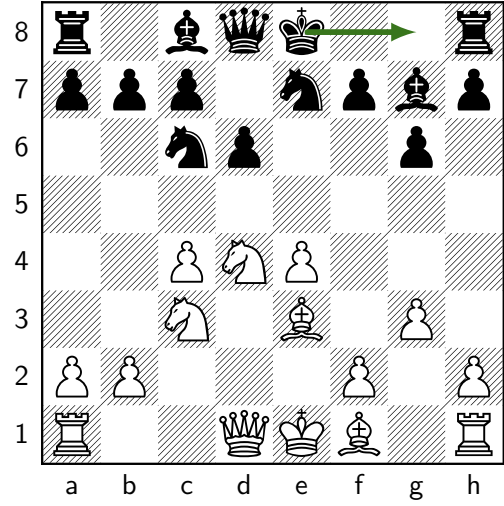


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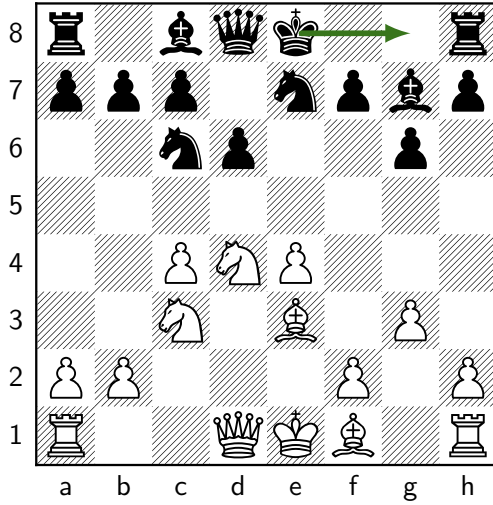
Figure 8: **Move 14, White: b3**. (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1799` (ply 19) [structural distance  $d=88$ , QAP obj=24]: the master played g6; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pg2$ ; cross-type roles: our  $Rf1$  as their  $Bc8$ , our  $Pf2$  as their  $Bf8$ ; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1800` (ply 19) [structural distance  $d=88$ , QAP obj=24]: the master played g6; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pg2$ ; cross-type roles: our  $Rf1$  as their  $Bc8$ , our  $Pf2$  as their  $Bf8$ ; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2474` (ply 19) [structural distance  $d=88$ , QAP obj=24]: the master played g6; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pg2$ ; cross-type roles: our  $Rf1$  as their  $Bc8$ , our  $Pf2$  as their  $Bf8$ ; transferred move b2b3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to b3 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



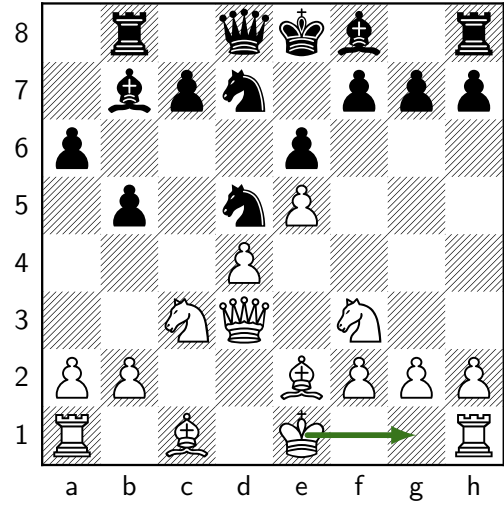
(a)



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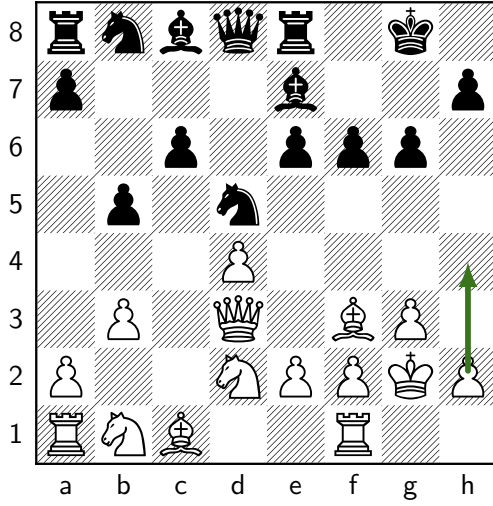


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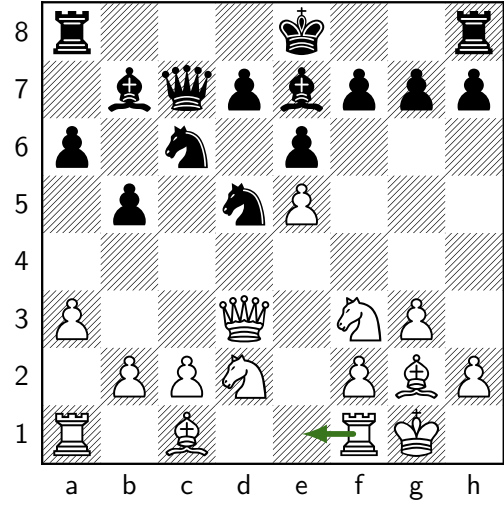


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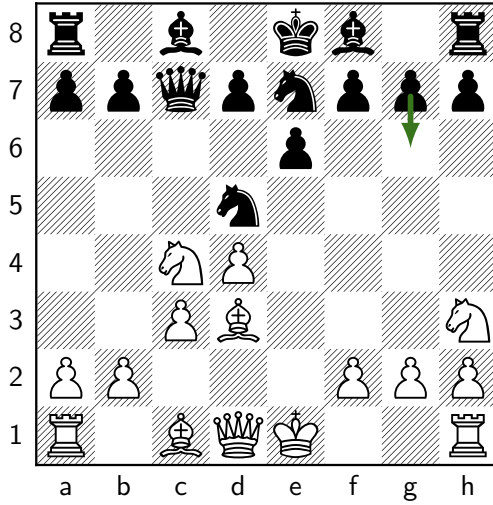
Figure 9: **Move 15, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1597` (ply 15) [structural distance  $d=79$ , QAP obj=35]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nd2* as their *Ra8*, our *Pb3* as their *Bc8*; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#1598` (ply 15) [structural distance  $d=79$ , QAP obj=35]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nd2* as their *Ra8*, our *Pb3* as their *Bc8*; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#1342` (ply 20) [structural distance  $d=98$ , QAP obj=34]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Ra1*, our *pe6* as their *Bc1*; transferred move **g1g2 (endorses this move)**. Synthesis: 9 retrieved analogs transferred to Kg2 (summed QAP objective 223, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



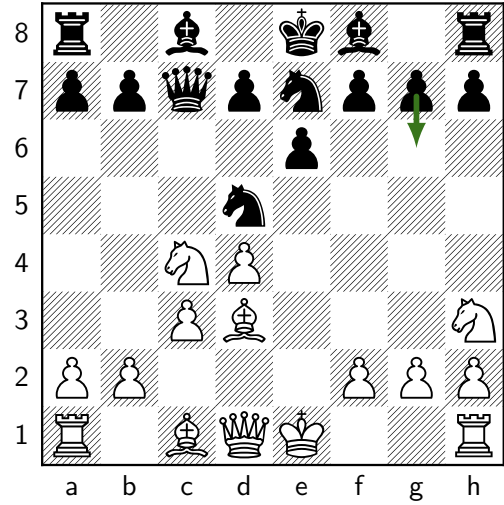
(a)



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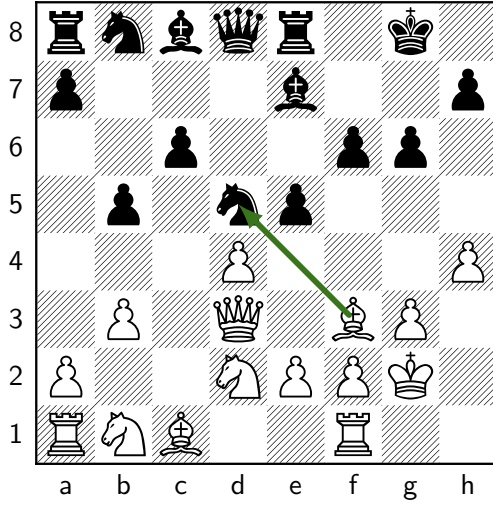


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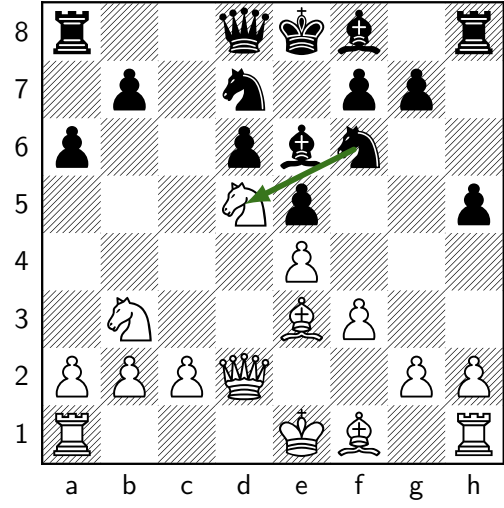


(d)

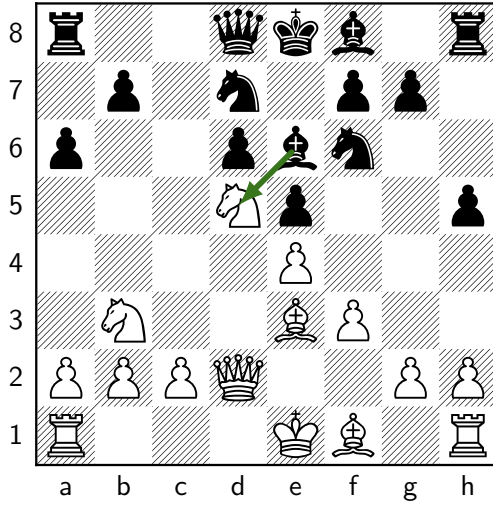
Figure 10: **Move 16, White: h4.** (a) Our position; Melanie plays h4 (h2h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3349` (ply 22) [structural distance  $d=91$ , QAP obj=34]: the master played  $\text{Re1}$ ; under  $\mu$  our  $\text{Ph2}$  plays the role of that game’s  $\text{Rf1}$  (a **cross-type** correspondence); cross-type roles: our  $\text{Pa2}$  as their  $\text{Nd2}$ , our  $\text{Nb1}$  as their  $\text{Pf2}$ ; transferred move **h2h4 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#1799` (ply 19) [structural distance  $d=99$ , QAP obj=24]: the master played  $\text{g6}$ ; under  $\mu$  our  $\text{Ph2}$  plays the role of that game’s  $\text{Pg2}$ ; cross-type roles: our  $\text{pc6}$  as their  $\text{Ra8}$ , our  $\text{ra8}$  as their  $\text{Bc8}$ ; transferred move **h2h4 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#1800` (ply 19) [structural distance  $d=99$ , QAP obj=24]: the master played  $\text{g6}$ ; under  $\mu$  our  $\text{Ph2}$  plays the role of that game’s  $\text{Pg2}$ ; cross-type roles: our  $\text{pc6}$  as their  $\text{Ra8}$ , our  $\text{ra8}$  as their  $\text{Bc8}$ ; transferred move **h2h4 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to h4 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



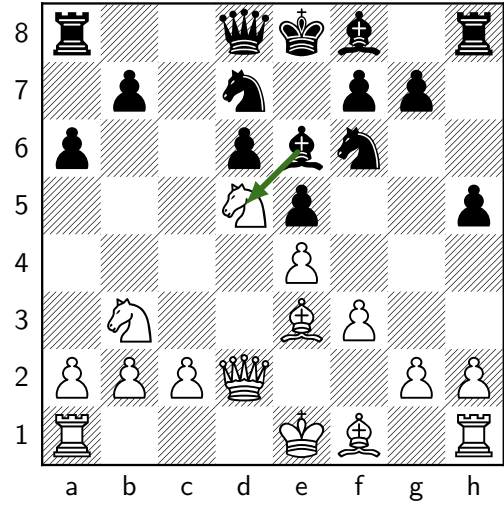
(a)



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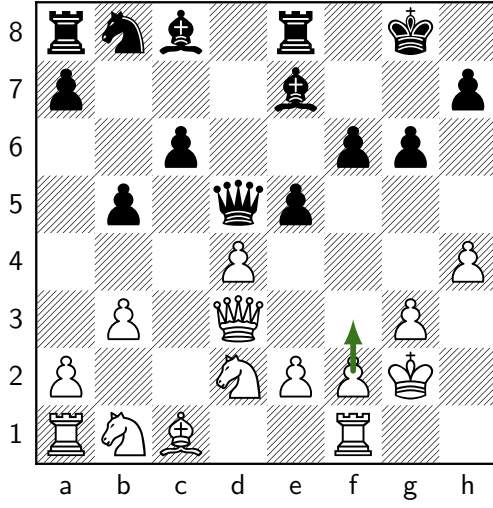


(c)

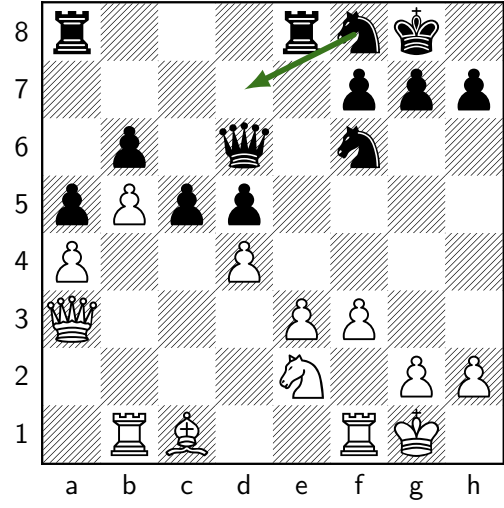


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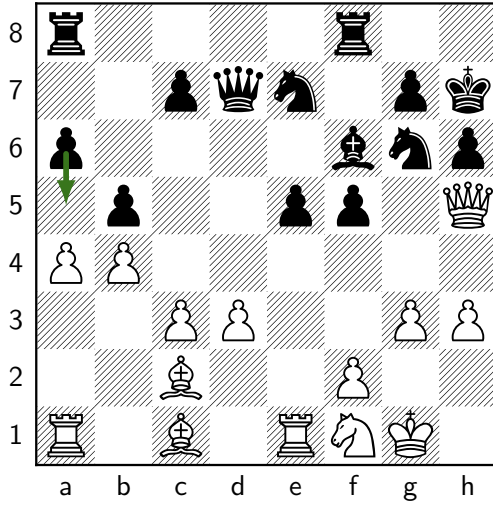
Figure 11: **Move 17, White: Bxd5+.** (a) Our position; Melanie plays Bxd5+ (f3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#406` (ply 19) [structural distance  $d=98$ , QAP obj=25]: the master played Nxd5; under  $\mu$  our Bf3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our be7 as their Ra8, our Nd2 as their Qd8; transferred move f3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#850` (ply 19) [structural distance  $d=98$ , QAP obj=25]: the master played Bxd5; under  $\mu$  our Bf3 plays the role of that game’s Be3; cross-type roles: our be7 as their Ra8, our Nd2 as their Qd8; transferred move f3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#851` (ply 19) [structural distance  $d=98$ , QAP obj=25]: the master played Bxd5; under  $\mu$  our Bf3 plays the role of that game’s Be3; cross-type roles: our be7 as their Ra8, our Nd2 as their Qd8; transferred move f3d5 (**endorses this move**). Synthesis: 25 retrieved analogs transferred to Bxd5+ (summed QAP objective 623, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



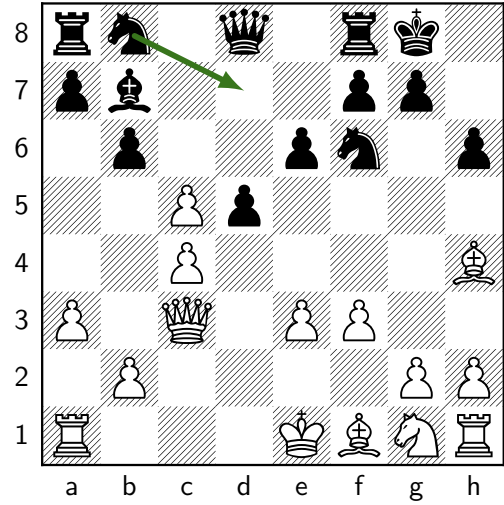
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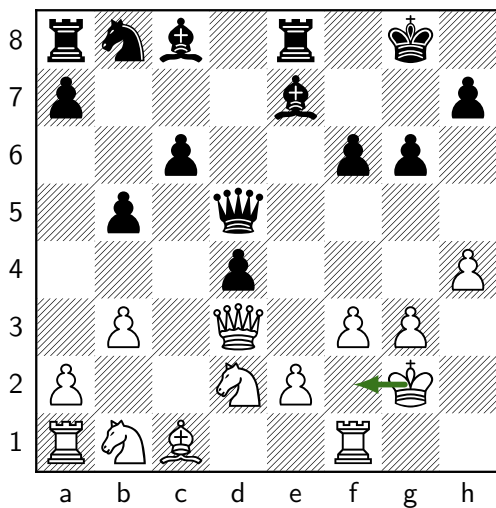


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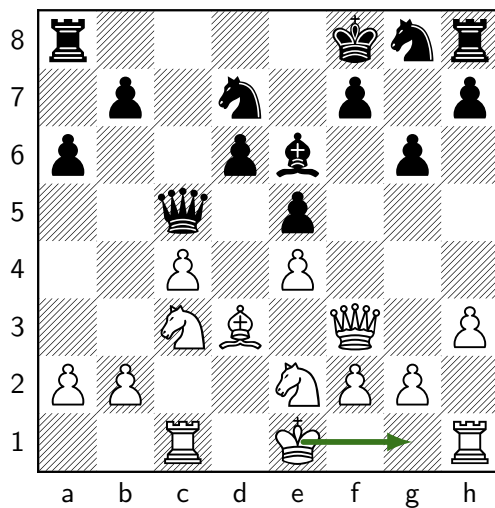


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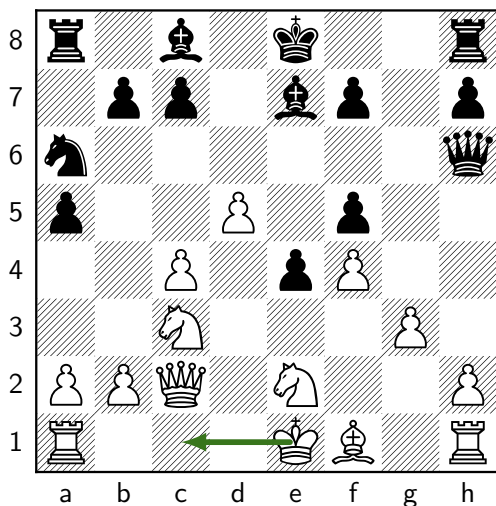
Figure 12: **Move 18, White: f3.** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#868` (ply 37) [structural distance  $d=104$ , QAP obj=36]: the master played N8d7; under  $\mu$  our Pf2 plays the role of that game’s Nf1 (a **cross-type** correspondence); cross-type roles: our Nd2 as their Ra8, our Ra1 as their Pg7; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#746` (ply 47) [structural distance  $d=110$ , QAP obj=28]: the master played a5; under  $\mu$  our Pf2 plays the role of that game’s Pa3; cross-type roles: our pc6 as their Ra8, our Bc1 as their Rf8; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1598` (ply 21) [structural distance  $d=104$ , QAP obj=19]: the master played Nbd7; under  $\mu$  our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra8, our Pg3 as their Rf8; transferred move f2f3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f3 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



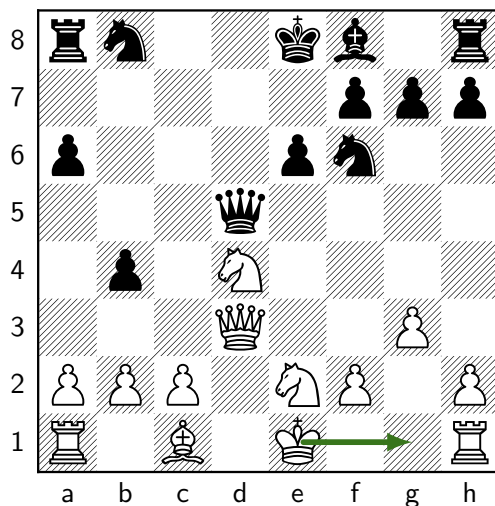
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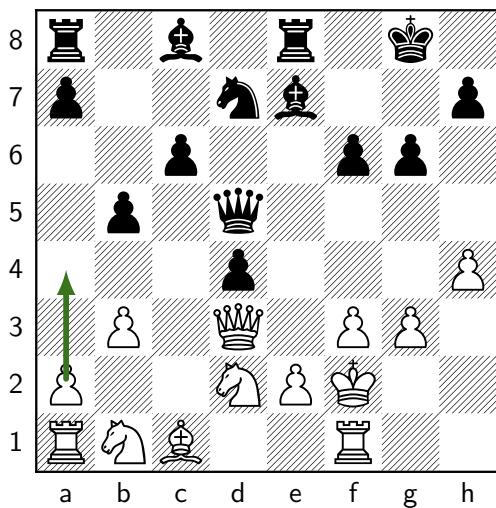


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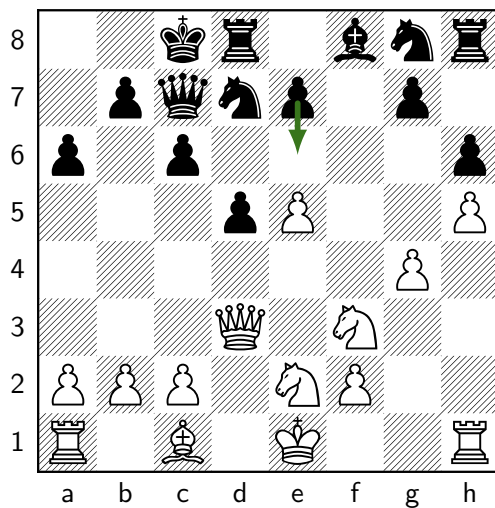


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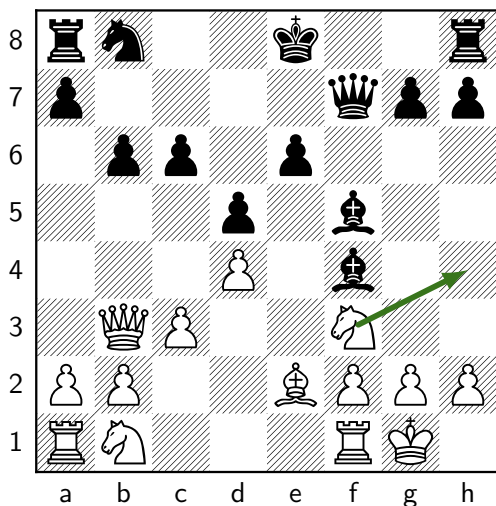
Figure 13: **Move 19, White: Kf2.** (a) Our position; Melanie plays Kf2 (g2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2284` (ply 28) [structural distance  $d=108$ , QAP obj=91]: the master played 0-0; under  $\mu$  our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *pa7* as their *Rc1*, our *Nd2* as their *Rh1*; transferred move g2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#802` (ply 26) [structural distance  $d=88$ , QAP obj=54]: the master played 0-0-0; under  $\mu$  our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *Pe2* as their *Bf1*, our *be7* as their *Rh1*; transferred move g2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2803` (ply 22) [structural distance  $d=108$ , QAP obj=24]: the master played 0-0; under  $\mu$  our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Ra1*, our *pc6* as their *Bc1*; transferred move g2f2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf2 (summed QAP objective 192, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



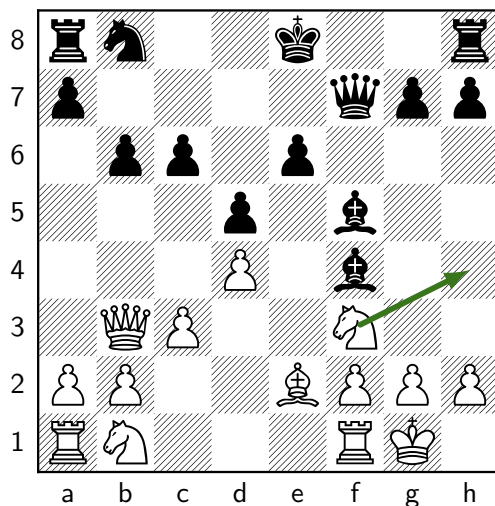
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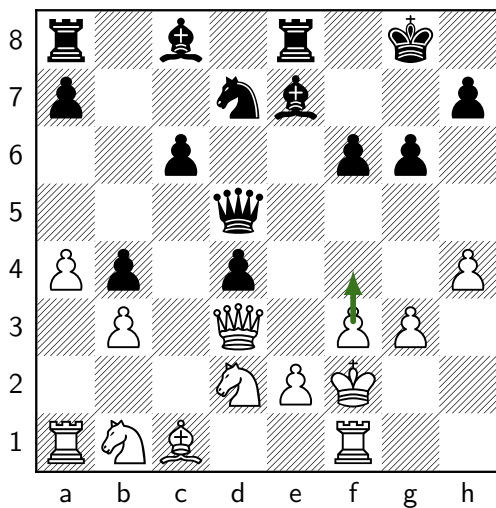
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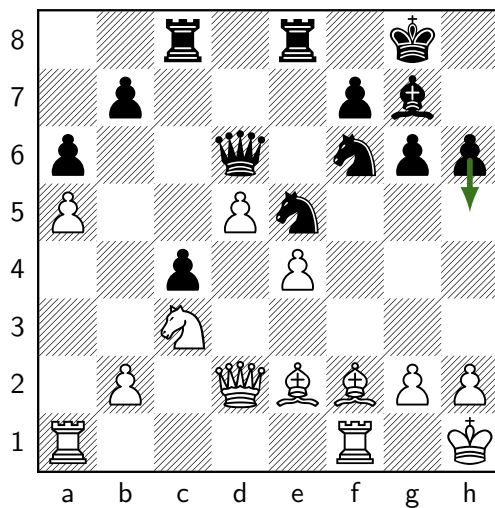
(d)

Figure 14: **Move 20, White: a4.** (a) Our position; Melanie plays a4 (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#929 (ply 25) [structural distance  $d=104$ , QAP obj=60]: the master played e6; under  $\mu$  our Pa2 plays the role of that game’s Pe2; cross-type roles: our Ra1 as their Bf8, our Nd2 as their Rh8; transferred move a2a4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#2422 (ply 26) [structural distance  $d=96$ , QAP obj=54]: the master played Nh4; under  $\mu$  our Pa2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our pd4 as their Nb1, our ra8 as their Be2; transferred move a2a4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#2102 (ply 26) [structural distance  $d=96$ , QAP obj=54]: the master played Nh4; under  $\mu$  our Pa2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our pd4 as their Nb1, our ra8 as their Be2; transferred move a2a4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to a4 (summed QAP objective 291, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

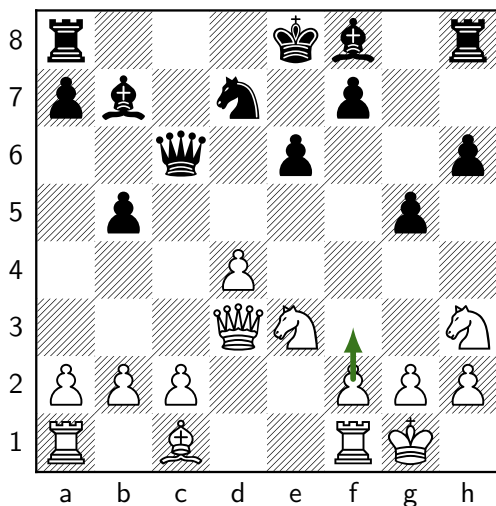




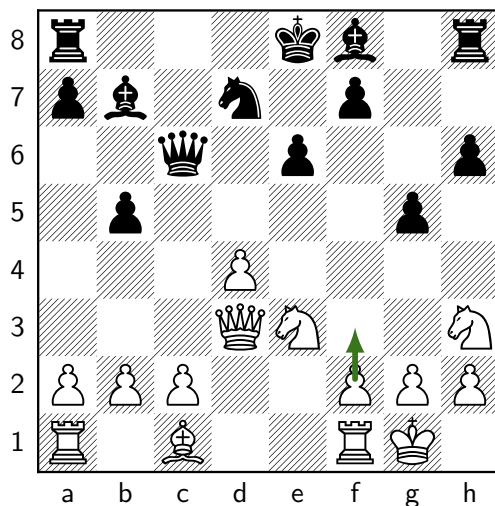
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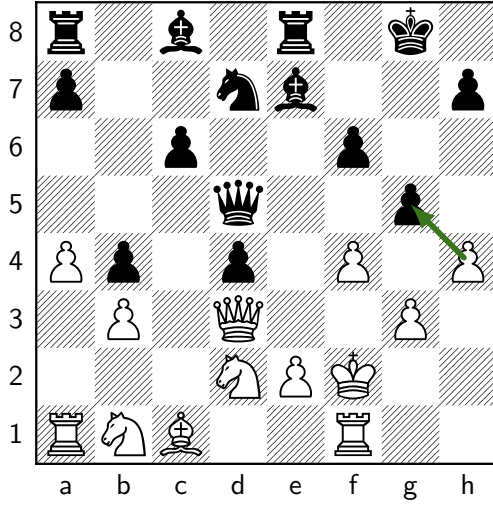
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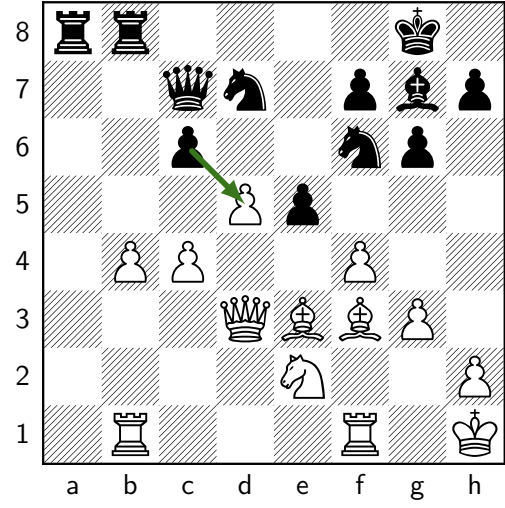
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Figure 15: **Move 21, White: f4**. (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2147` (ply 39) [structural distance  $d=110$ , QAP obj=73]: the master played h5; under  $\mu$  our  $Pf3$  plays the role of that game’s  $Ph3$ ; cross-type roles: our  $be7$  as their  $Re8$ , our  $Nb1$  as their  $Pb7$ ; transferred move f3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#637` (ply 34) [structural distance  $d=100$ , QAP obj=46]: the master played f3; under  $\mu$  our  $Pf3$  plays the role of that game’s  $Pf2$ ; cross-type roles: our  $Nb1$  as their  $Ra1$ , our  $pc6$  as their  $Ne3$ ; transferred move f3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3526` (ply 34) [structural distance  $d=100$ , QAP obj=46]: the master played f3; under  $\mu$  our  $Pf3$  plays the role of that game’s  $Pf2$ ; cross-type roles: our  $Nb1$  as their  $Ra1$ , our  $pc6$  as their  $Ne3$ ; transferred move f3f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f4 (summed QAP objective 197, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

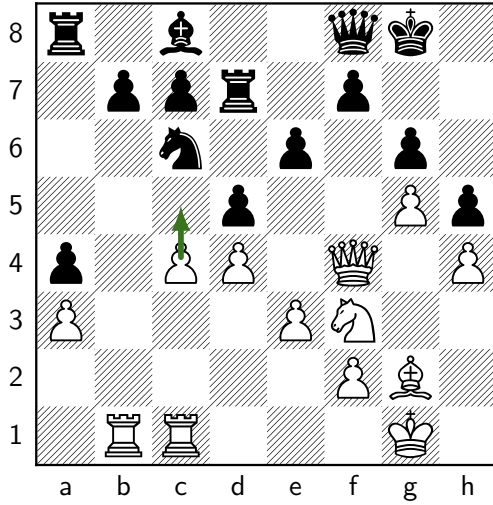




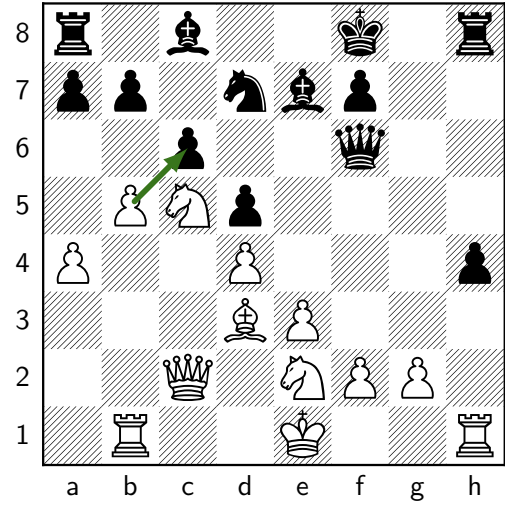
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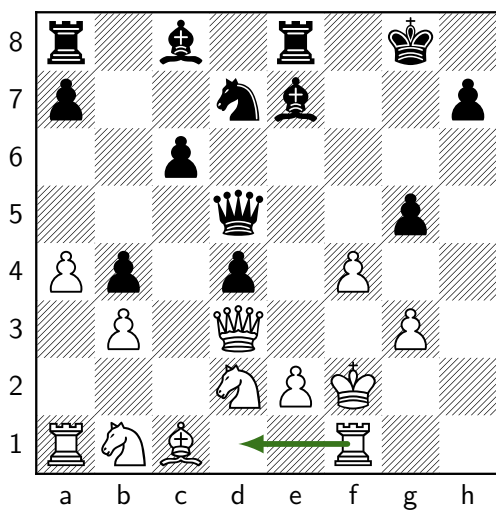


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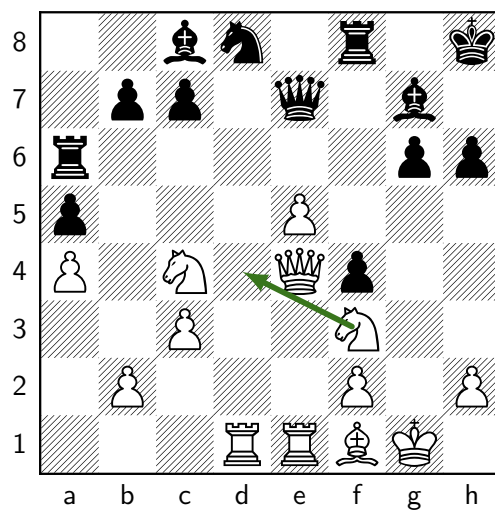


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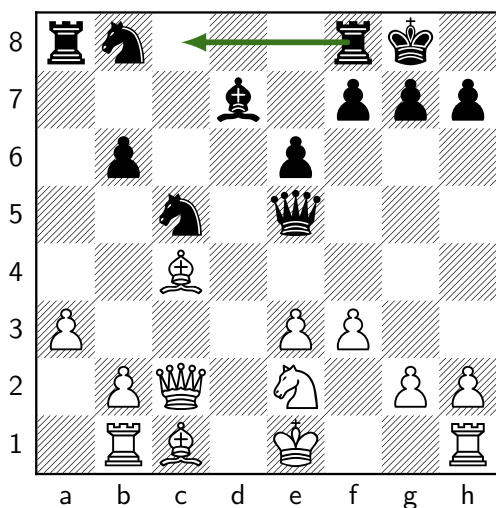
Figure 16: **Move 22, White: hxg5.** (a) Our position; Melanie plays **h4g5** (**h4g5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1139` (ply 45) [structural distance  $d=113$ , QAP obj=101]: the master played **cxd5**; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pc3$ ; cross-type roles: our  $Pe2$  as their  $Ra8$ , our  $Qd3$  as their  $Rb8$ ; transferred move **h4g5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#772` (ply 44) [structural distance  $d=114$ , QAP obj=81]: the master played **c5**; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pc4$ ; cross-type roles: our  $Pe2$  as their  $Rb1$ , our  $Nd2$  as their  $Bg2$ ; transferred move **h4g5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#518` (ply 44) [structural distance  $d=115$ , QAP obj=77]: the master played **bxc6**; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pb5$ ; cross-type roles: our  $Qd3$  as their  $Rh1$ , our  $Nd2$  as their  $Qc2$ ; transferred move **h4g5** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to **hxg5** (summed QAP objective 451, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



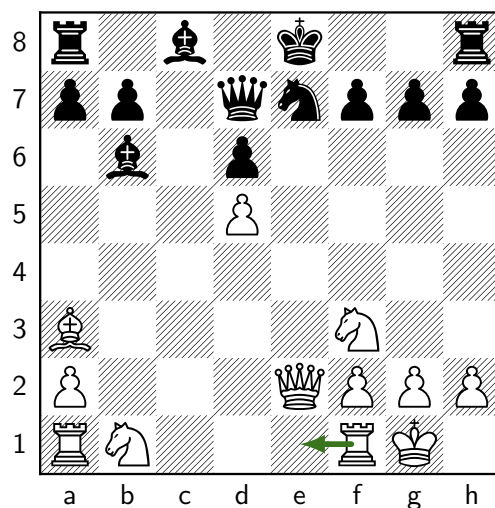
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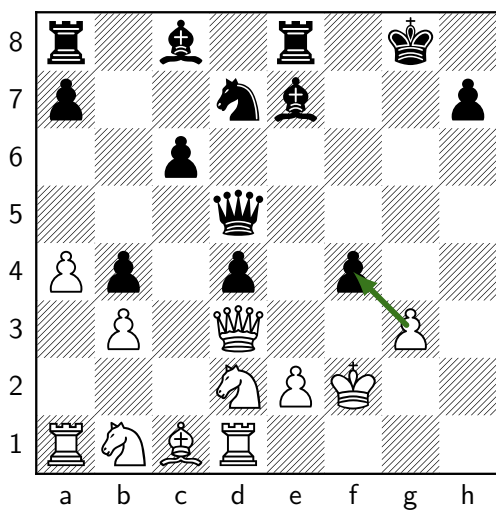


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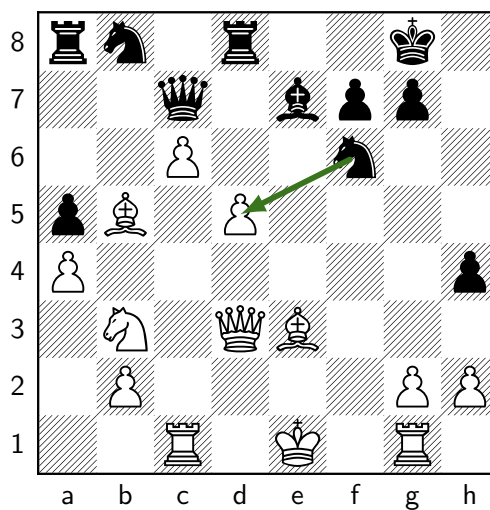


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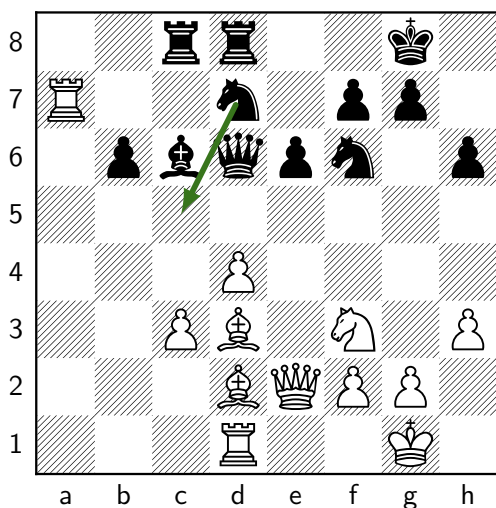
Figure 17: **Move 23, White: Rd1.** (a) Our position; Melanie plays Rd1 (f1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#38` (ply 52) [structural distance  $d=107$ , QAP obj=68]: the master played Nd4; under  $\mu$  our *Rf1* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *pc6* as their *Rd1*, our *be7* as their *Re1*; transferred move f1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2507` (ply 29) [structural distance  $d=78$ , QAP obj=64]: the master played Rc8; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pf4* as their *Bd7*, our *Nb1* as their *Ph7*; transferred move f1d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#24` (ply 32) [structural distance  $d=109$ , QAP obj=43]: the master played Re1; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb1* as their *Ra1*, our *be7* as their *Nb1*; transferred move f1d1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rd1 (summed QAP objective 292, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



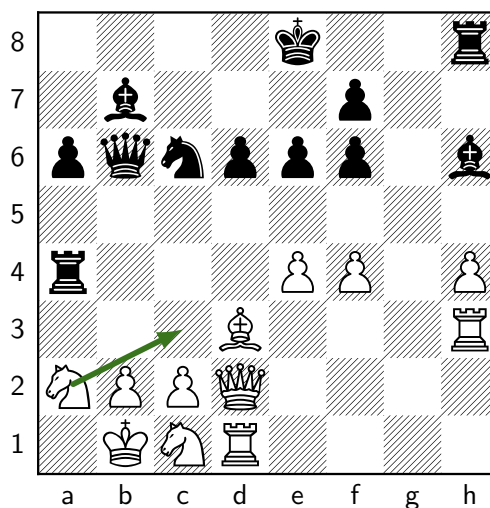
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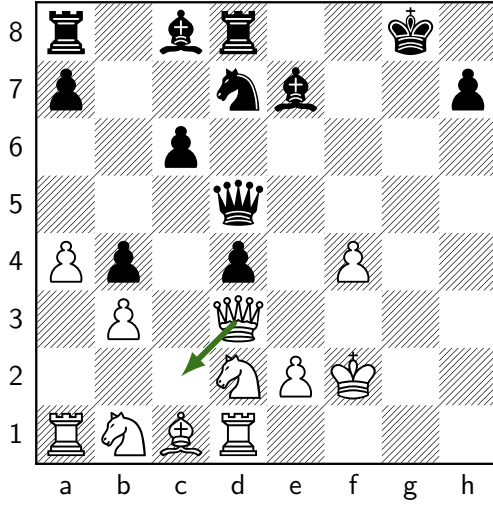


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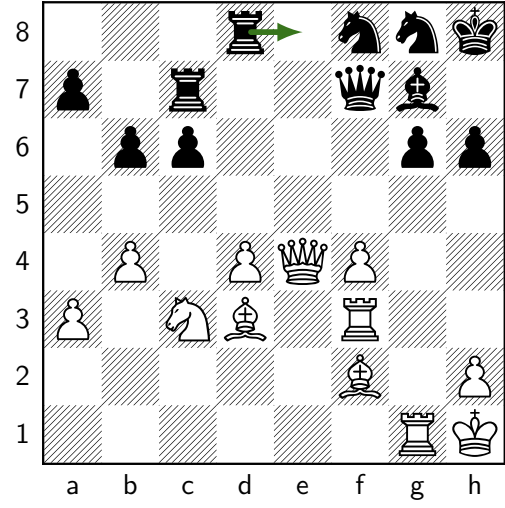


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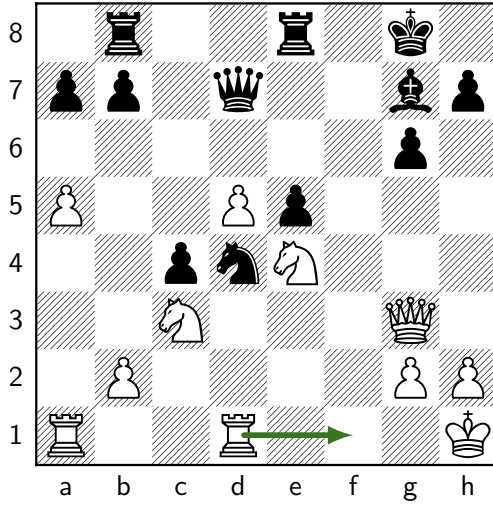
Figure 18: **Move 24, White: gxf4.** (a) Our position; Melanie plays gxf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2193` (ply 45) [structural distance  $d=143$ , QAP obj=74]: the master played Nxd5; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Nf3$  (a **cross-type** correspondence); cross-type roles: our  $Qd3$  as their  $Nb8$ , our  $Pg3$  as their  $Rd8$ ; transferred move g3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1822` (ply 45) [structural distance  $d=152$ , QAP obj=60]: the master played Nc5; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Nd2$  (a **cross-type** correspondence); cross-type roles: our  $Nb1$  as their  $Rc8$ , our  $Nd2$  as their  $Rd8$ ; transferred move g3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#956` (ply 46) [structural distance  $d=134$ , QAP obj=32]: the master played Nc3; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Na2$  (a **cross-type** correspondence); cross-type roles: our  $bc8$  as their  $Nc1$ , our  $Bc1$  as their  $Pb2$ ; transferred move g3f4 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to gxf4 (summed QAP objective 374, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



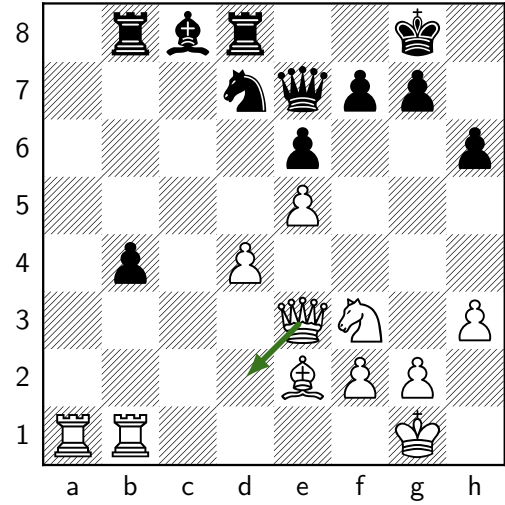
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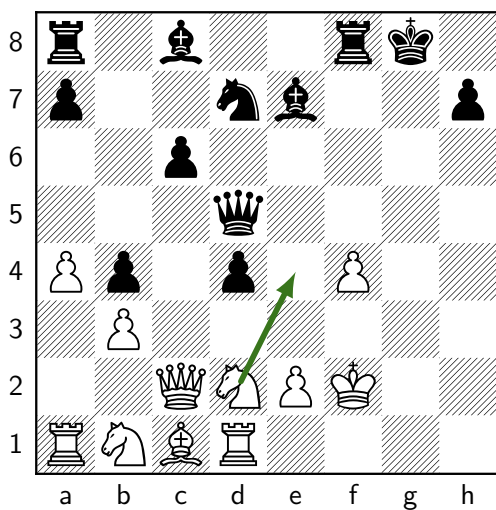


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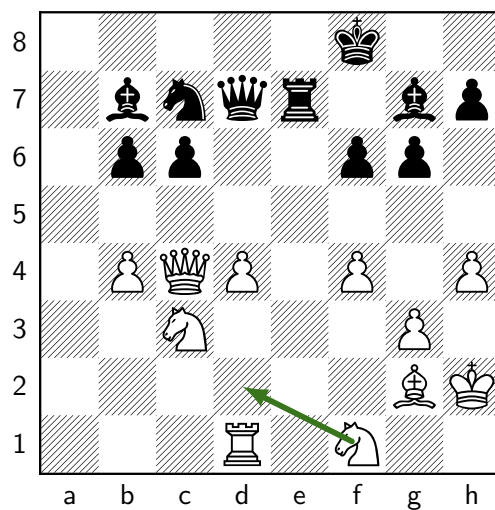


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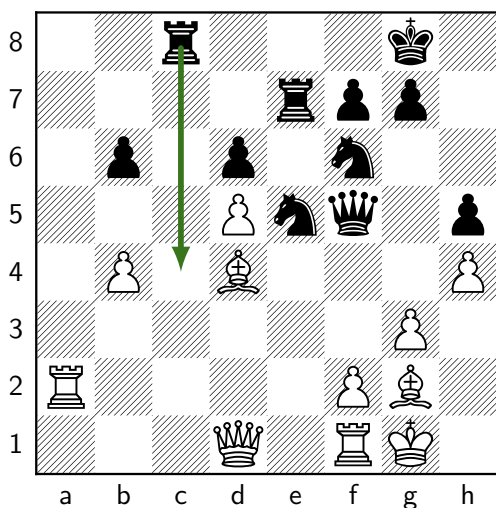
Figure 19: **Move 25, White: Qc2**. (a) Our position; Melanie plays Qc2 (d3c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3279` (ply 65) [structural distance  $d=138$ , QAP obj=57]: the master played Re8; under  $\mu$  our Qd3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Ng8, our Qd3 as their Bg7; transferred move d3c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1352` (ply 54) [structural distance  $d=134$ , QAP obj=54]: the master played Rf1; under  $\mu$  our Qd3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our Bc1 as their Pb2; transferred move d3c2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3221` (ply 46) [structural distance  $d=138$ , QAP obj=29]: the master played Qd2; under  $\mu$  our Qd3 plays the role of that game’s Qe3; cross-type roles: our Pe2 as their Ra1, our Qd3 as their Rb1; transferred move d3c2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc2 (summed QAP objective 195, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



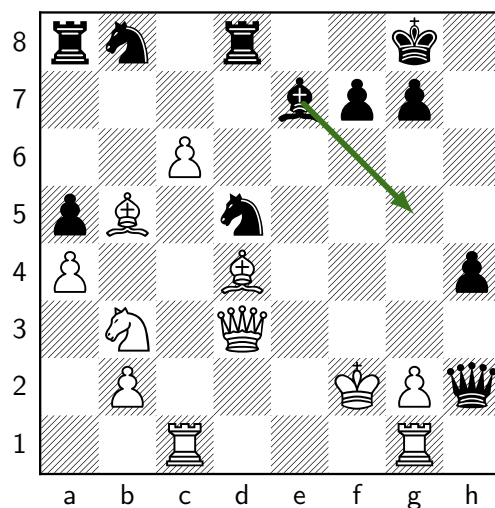
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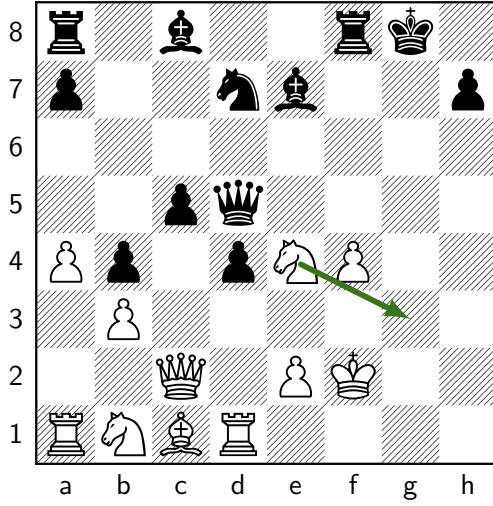


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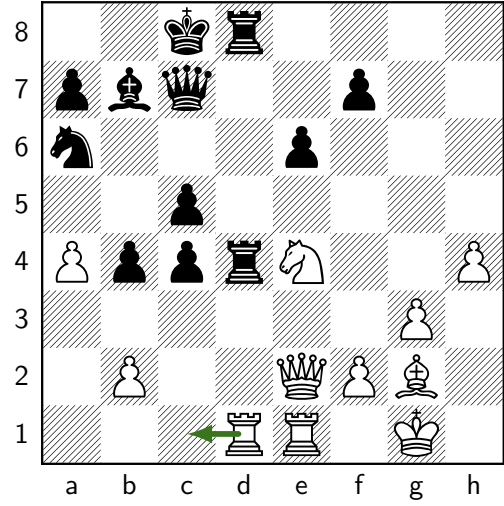


(d)

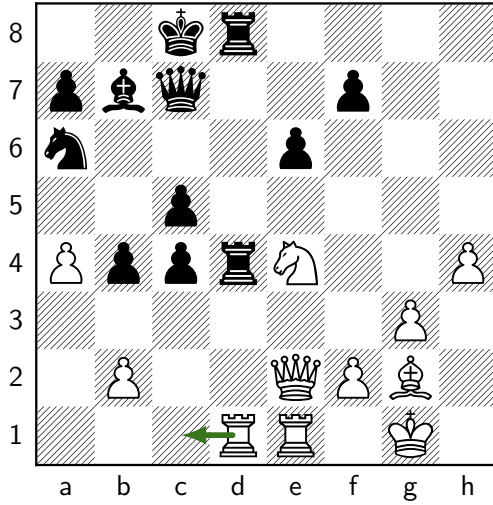
Figure 20: **Move 26, White: Ne4.** (a) Our position; Melanie plays Ne4 (d2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1988` (ply 62) [structural distance  $d=105$ , QAP obj=62]: the master played Nd2; under  $\mu$  our Nd2 plays the role of that game's Nf1; cross-type roles: our Qc2 as their Bg2, our Bc1 as their Nc3; transferred move d2e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1762` (ply 51) [structural distance  $d=118$ , QAP obj=24]: the master played Rc4; under  $\mu$  our Nd2 plays the role of that game's Rc1 (a **cross-type** correspondence); cross-type roles: our Pf4 as their Re7, our Pb3 as their Nf6; transferred move d2e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2193` (ply 49) [structural distance  $d=111$ , QAP obj=11]: the master played Bg5; under  $\mu$  our Nd2 plays the role of that game's Be2 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Rd8, our Bc1 as their Pa5; transferred move d2e4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne4 (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



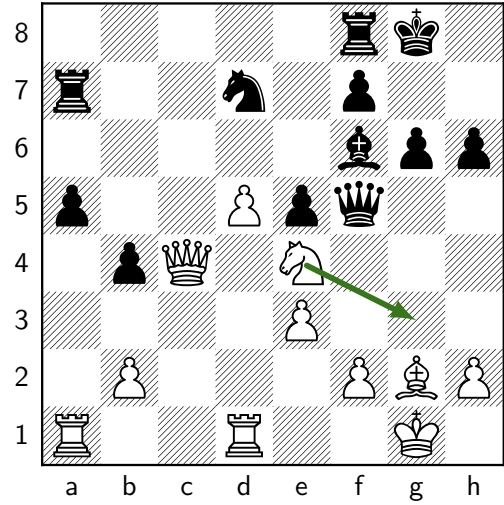
(a)



(b)

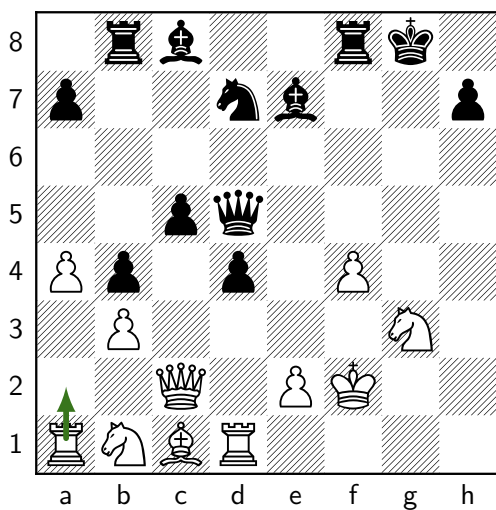


(c)

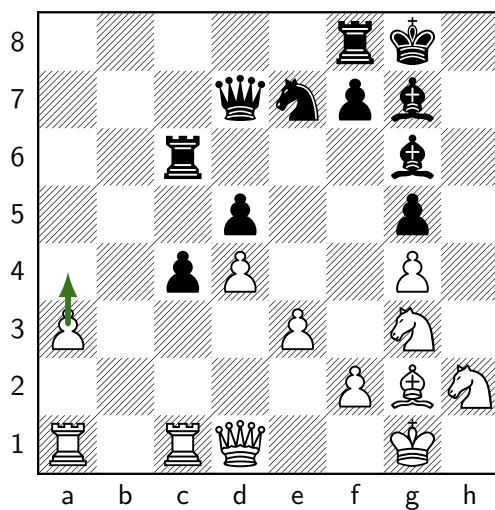


(d)

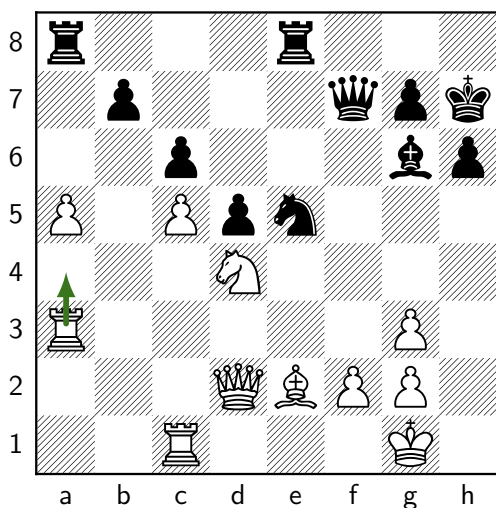
Figure 21: **Move 27, White: Ng3.** (a) Our position; Melanie plays Ng3 (e4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1545` (ply 42) [structural distance  $d=91$ , QAP obj=53]: the master played Rc1; under  $\mu$  our Ne4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Re1, our Nb1 as their Pf2; transferred move e4g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2138` (ply 42) [structural distance  $d=91$ , QAP obj=53]: the master played Rc1; under  $\mu$  our Ne4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Re1, our Nb1 as their Pf2; transferred move e4g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2173` (ply 44) [structural distance  $d=95$ , QAP obj=45]: the master played Ng3; under  $\mu$  our Ne4 plays the role of that game’s Ne4; cross-type roles: our be7 as their Rd1, our Ra1 as their Bg2; transferred move e4g3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ng3 (summed QAP objective 279, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



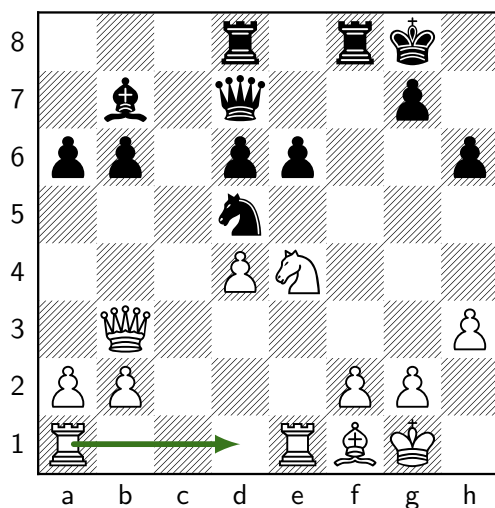
(a)



(b)



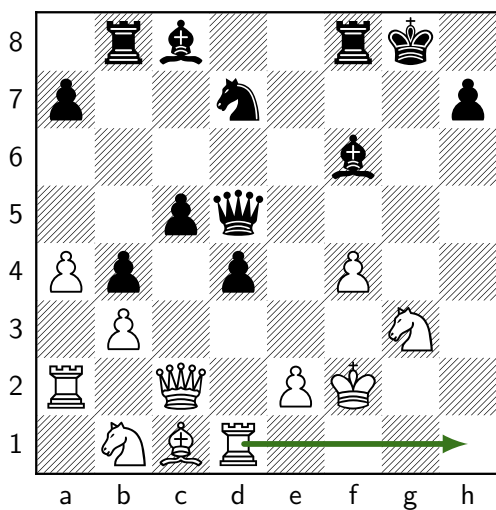
(c)



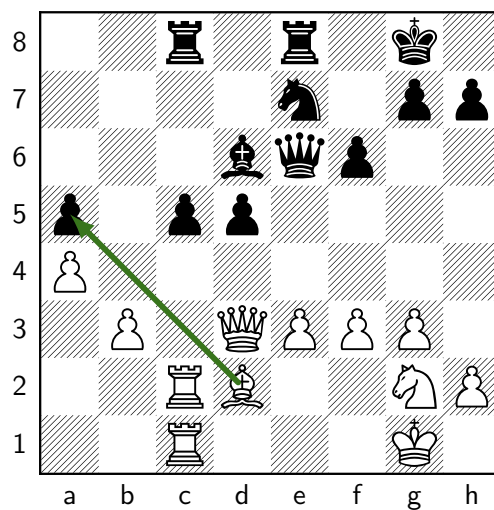
(d)

Figure 22: **Move 28, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1535` (ply 54) [structural distance  $d=86$ , QAP obj=106]: the master played a4; under  $\mu$  our Ra1 plays the role of that game's Pa3 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Ra1, our Qc2 as their Rc1; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#56` (ply 64) [structural distance  $d=75$ , QAP obj=71]: the master played Ra4; under  $\mu$  our Ra1 plays the role of that game's Ra3; cross-type roles: our Rd1 as their Qd2, our nd7 as their Be2; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1510` (ply 42) [structural distance  $d=89$ , QAP obj=48]: the master played Rad1; under  $\mu$  our Ra1 plays the role of that game's Ra1; cross-type roles: our ph7 as their Re1, our rb8 as their Bf1; transferred move a1a2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ra2 (summed QAP objective 247, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

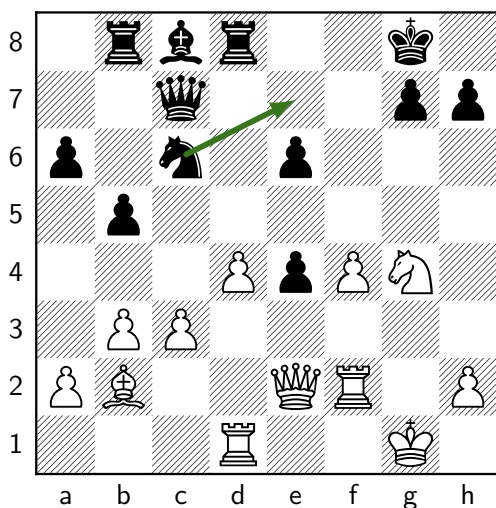




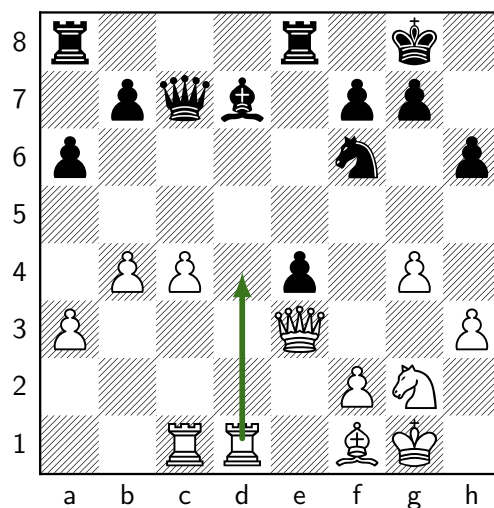
(a)



(b)

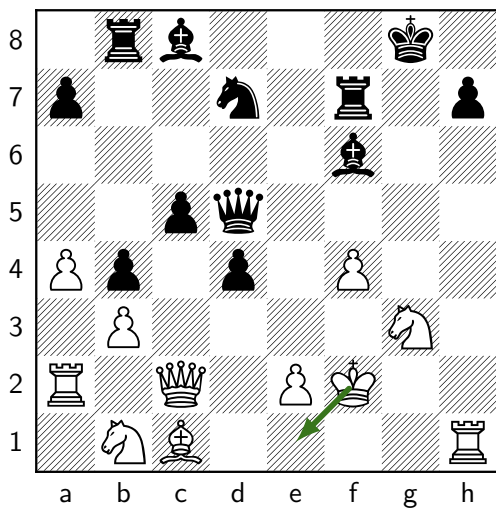


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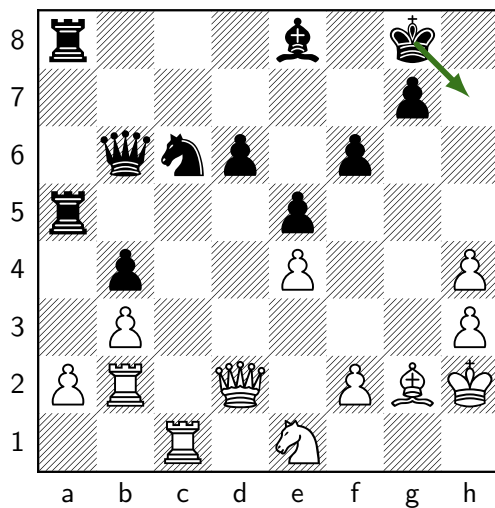


(d)

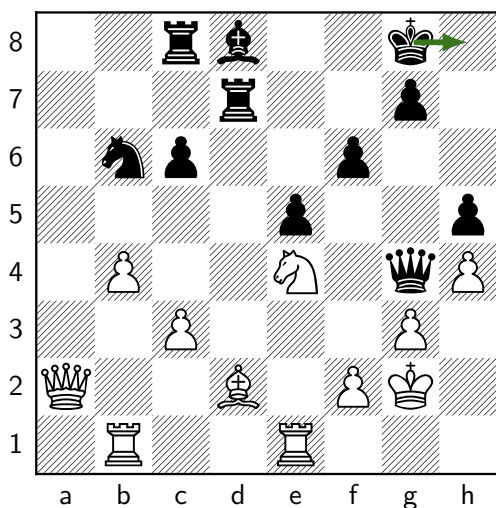
Figure 23: **Move 29, White: Rh1**. (a) Our position; Melanie plays Rh1 (d1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1326` (ply 68) [structural distance  $d=97$ , QAP obj=106]: the master played Bxa5; under  $\mu$  our *Rd1* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *Qc2* as their *Rc2*, our *Pe2* as their *Ng2*; transferred move d1h1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2292` (ply 43) [structural distance  $d=97$ , QAP obj=72]: the master played Ne7; under  $\mu$  our *Rd1* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Rb8*, our *Ra2* as their *Bc8*; transferred move d1h1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2270` (ply 64) [structural distance  $d=89$ , QAP obj=51]: the master played Rd4; under  $\mu$  our *Rd1* plays the role of that game’s *Rd1*; cross-type roles: our *Bc1* as their *Rc1*, our *Ra2* as their *Pf2*; transferred move d1h1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rh1 (summed QAP objective 236, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



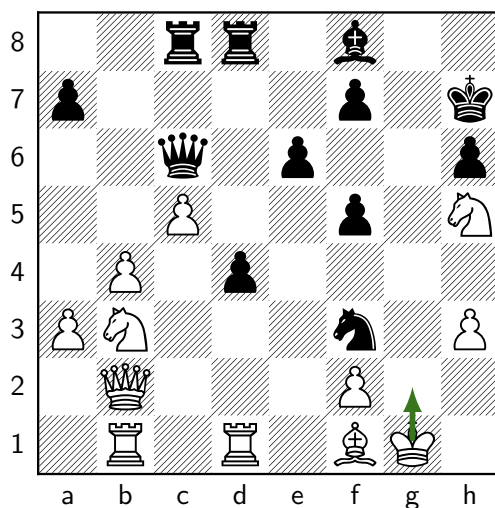
(a)



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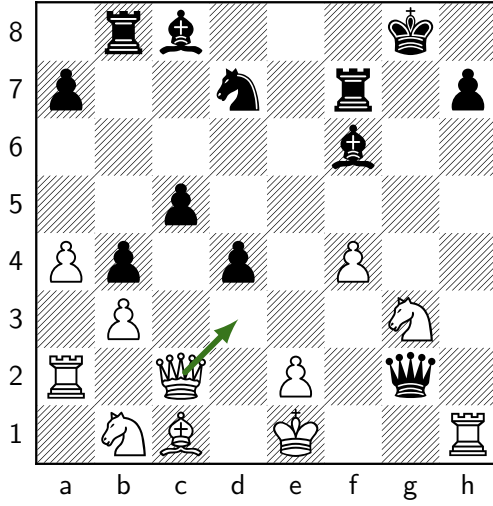


(c)

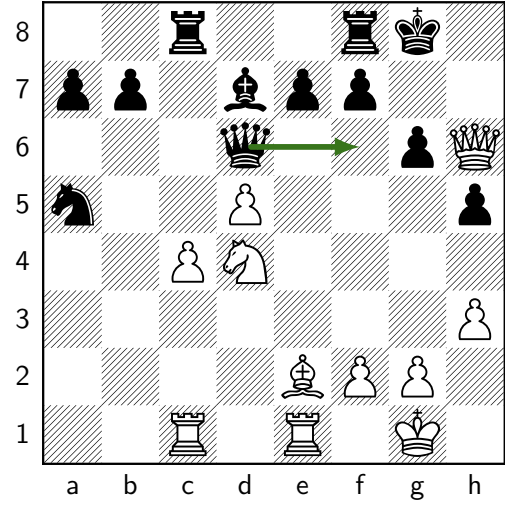


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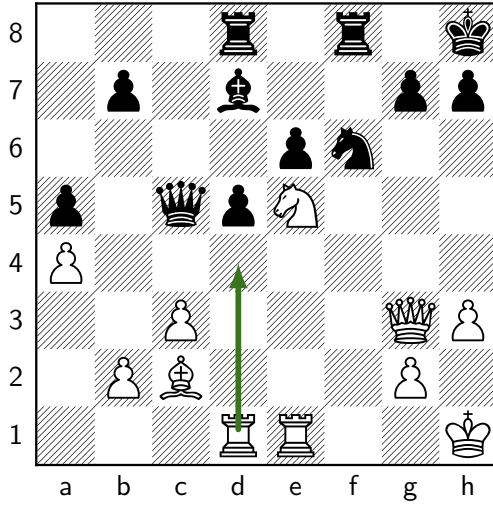
Figure 24: **Move 30, White: Ke1**. (a) Our position; Melanie plays Ke1 (f2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1629` (ply 63) [structural distance  $d=83$ , QAP obj=46]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game's Kg1; cross-type roles: our Pa4 as their Be8, our Rh1 as their Pd6; transferred move f2e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#340` (ply 65) [structural distance  $d=82$ , QAP obj=36]: the master played Kh8; under  $\mu$  our Kf2 plays the role of that game's Kg1; cross-type roles: our nd7 as their Rc8, our Ng3 as their Pg7; transferred move f2e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3196` (ply 64) [structural distance  $d=86$ , QAP obj=24]: the master played Kg2; under  $\mu$  our Kf2 plays the role of that game's Kg1; cross-type roles: our Qc2 as their Rd1, our Ng3 as their Bf1; transferred move f2e1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke1 (summed QAP objective 131, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



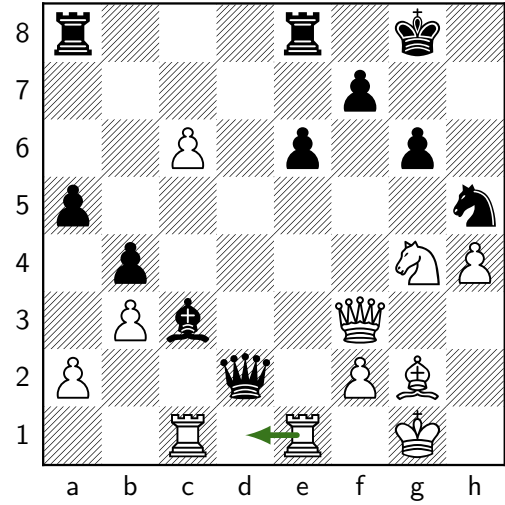
(a)



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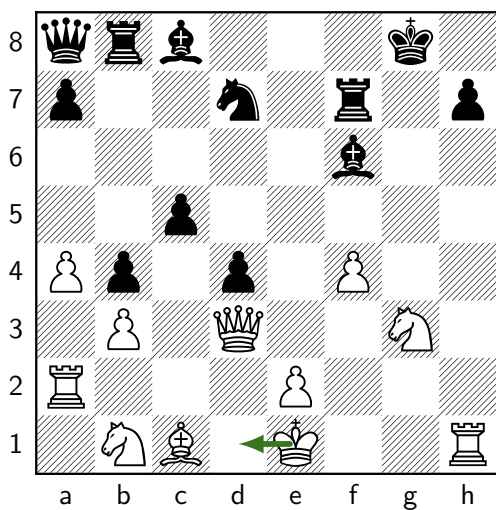


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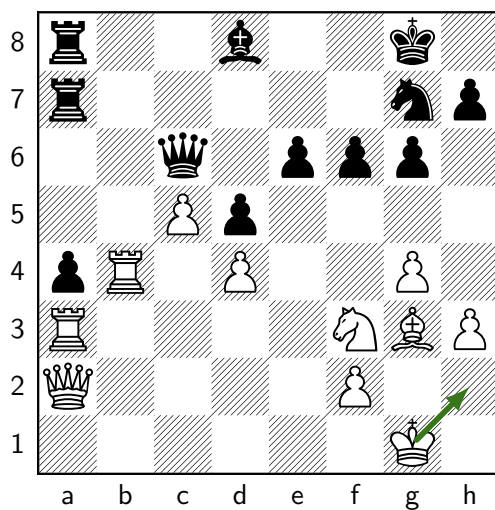


(d)

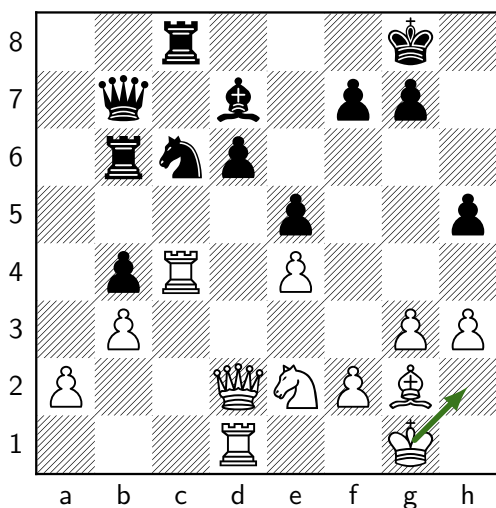
Figure 25: **Move 31, White: Qd3.** (a) Our position; Melanie plays Qd3 (c2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#187` (ply 45) [structural distance  $d=95$ , QAP obj=60]: the master played Qf6; under  $\mu$  our Qc2 plays the role of that game’s Qd3; cross-type roles: our nd7 as their Rc8, our Nb1 as their Pa7; transferred move c2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3051` (ply 60) [structural distance  $d=98$ , QAP obj=48]: the master played Rd4; under  $\mu$  our Qc2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Re1, our Nb1 as their Pb2; transferred move c2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3904` (ply 60) [structural distance  $d=90$ , QAP obj=85]: the master played Red1; under  $\mu$  our Ng3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our nd7 as their Bg2, our pb4 as their bc3; transferred move g3e4. Synthesis: 2 retrieved analogs transferred to Qd3 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



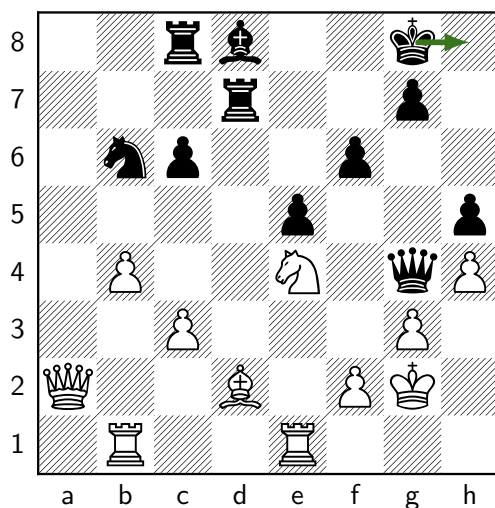
(a)



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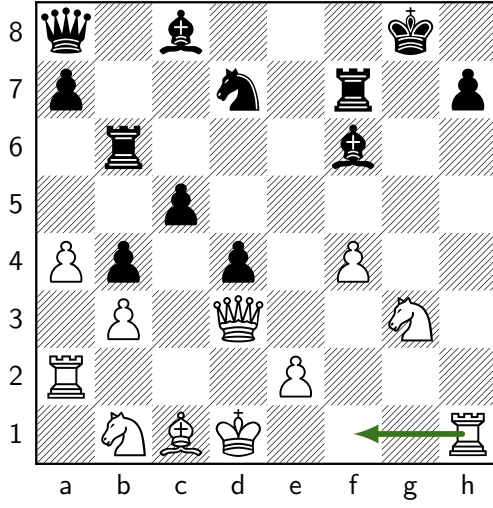


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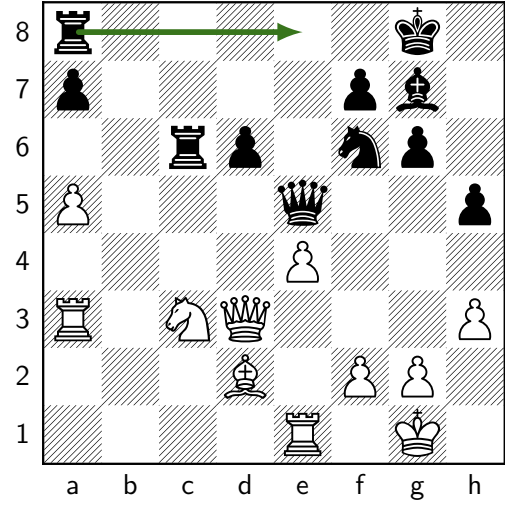


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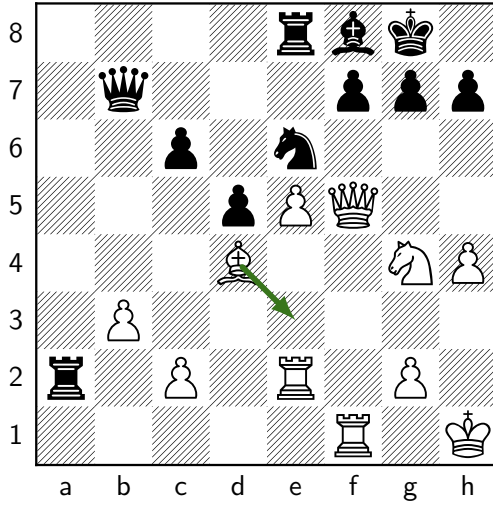
Figure 26: **Move 32, White: Kd1.** (a) Our position; Melanie plays Kd1 (e1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2949` (ply 70) [structural distance  $d=94$ , QAP obj=72]: the master played Kh2; under  $\mu$  our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *Qd3* as their *Ra3*, our *Pb3* as their *Nf3*; transferred move e1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1629` (ply 48) [structural distance  $d=90$ , QAP obj=45]: the master played Kh2; under  $\mu$  our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *Nb1* as their *Pa2*, our *Pb3* as their *Ne2*; transferred move e1d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#340` (ply 65) [structural distance  $d=89$ , QAP obj=30]: the master played Kh8; under  $\mu$  our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *Pb3* as their *Bd8*, our *bc8* as their *Rd7*; transferred move e1d1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd1 (summed QAP objective 219, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



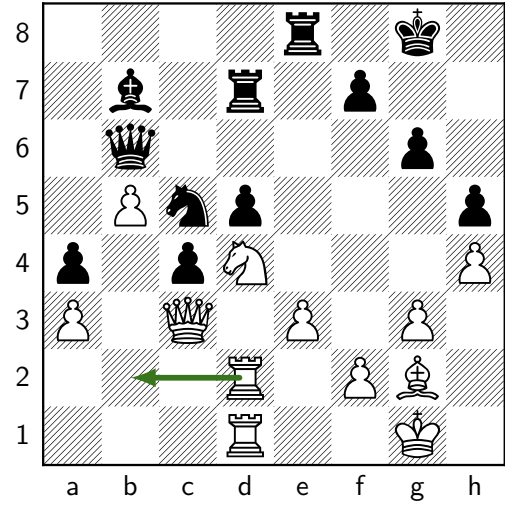
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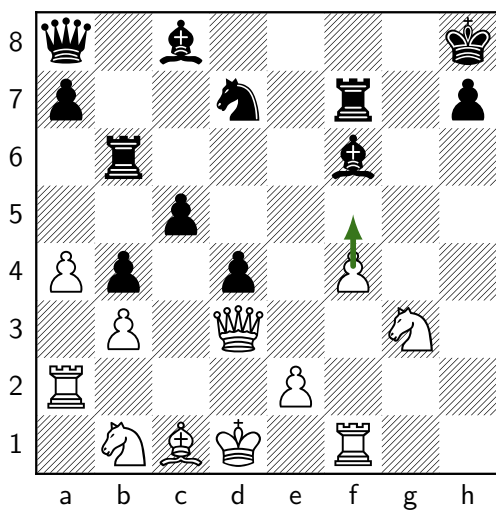


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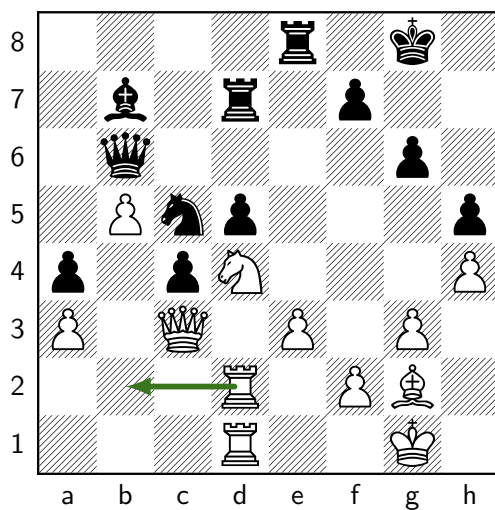


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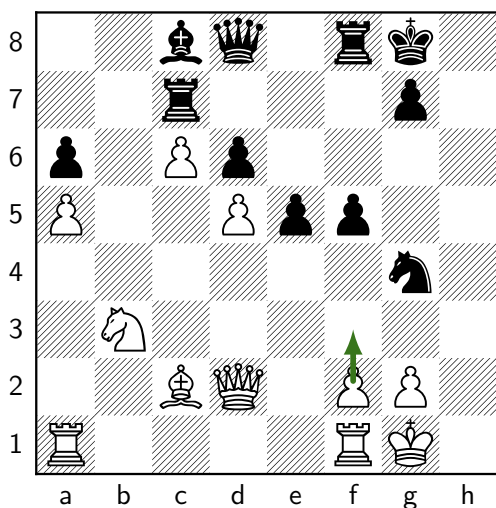
Figure 27: **Move 33, White: Rf1.** (a) Our position; Melanie plays Rf1 (h1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#549` (ply 57) [structural distance  $d=97$ , QAP obj=65]: the master played R e8; under  $\mu$  our *Rh1* plays the role of that game’s *Ra1*; cross-type roles: our *Bc1* as their *Pa7*, our *Ra2* as their *Pf7*; transferred move h1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#245` (ply 70) [structural distance  $d=100$ , QAP obj=23]: the master played B e3; under  $\mu$  our *Rh1* plays the role of that game’s *Bd4* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Rf1*, our *nd7* as their *ra2*; transferred move h1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2023` (ply 58) [structural distance  $d=100$ , QAP obj=61]: the master played R b2; under  $\mu$  our *Nb1* plays the role of that game’s *Rd2* (a **cross-type** correspondence); cross-type roles: our *pa7* as their *Rd1*, our *Qd3* as their *Bg2*; transferred move b1a3. Synthesis: 2 retrieved analogs transferred to Rf1 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



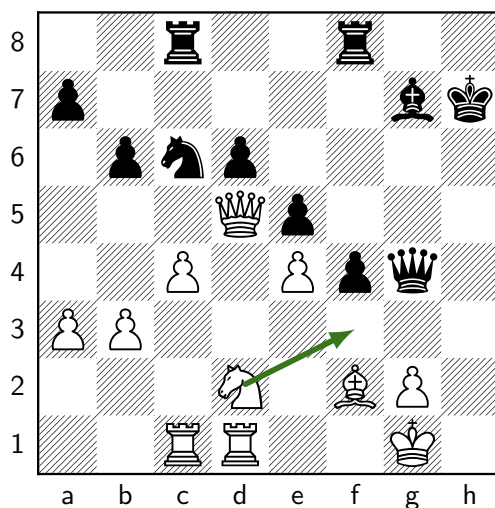
(a)



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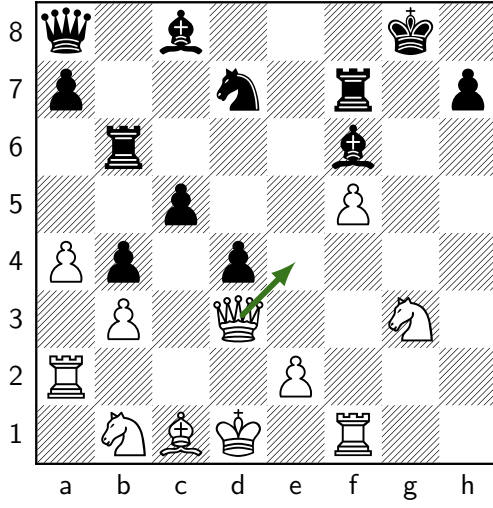


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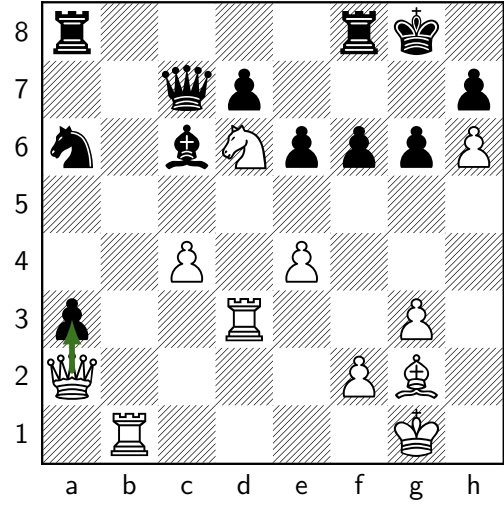


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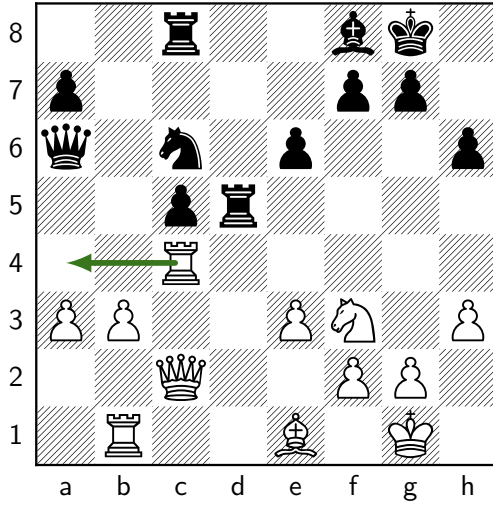
Figure 28: **Move 34, White: f5.** (a) Our position; Melanie plays f5 (f4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2023` (ply 58) [structural distance  $d=101$ , QAP obj=58]: the master played Rb2; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Rd2$  (a **cross-type** correspondence); cross-type roles: our  $Qd3$  as their  $Bg2$ , our  $Ra2$  as their  $Pa3$ ; transferred move f4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2875` (ply 52) [structural distance  $d=114$ , QAP obj=44]: the master played f3; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Pf_2$ ; cross-type roles: our  $Pb3$  as their  $Rf1$ , our  $Qd3$  as their  $Bc2$ ; transferred move f4f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2288` (ply 54) [structural distance  $d=106$ , QAP obj=17]: the master played Nf3; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Nd2$  (a **cross-type** correspondence); cross-type roles: our  $Ng3$  as their  $Rc1$ , our  $pd4$  as their  $Rd1$ ; transferred move f4f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f5 (summed QAP objective 136, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



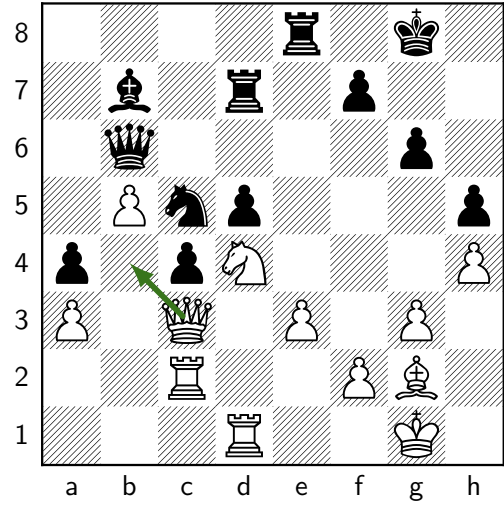
(a)



(b)



(c)



(d)

Figure 29: **Move 35, White: Qe4.** (a) Our position; Melanie plays Qe4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3605` (ply 54) [structural distance  $d=119$ , QAP obj=33]: the master played Qxa3; under  $\mu$  our Qd3 plays the role of that game’s Qa2; cross-type roles: our Nb1 as their Rb1, our rb6 as their Bg2; transferred move d3e4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#461` (ply 78) [structural distance  $d=116$ , QAP obj=32]: the master played Ra4; under  $\mu$  our Qd3 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Rb1, our Pe2 as their Be1; transferred move d3e4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2023` (ply 66) [structural distance  $d=104$ , QAP obj=57]: the master played Qb4; under  $\mu$  our Qd3 plays the role of that game’s Qc3; cross-type roles: our Pe2 as their Rc2, our pa7 as their Bg2; transferred move d3c4. Synthesis: 2 retrieved analogs transferred to Qe4 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



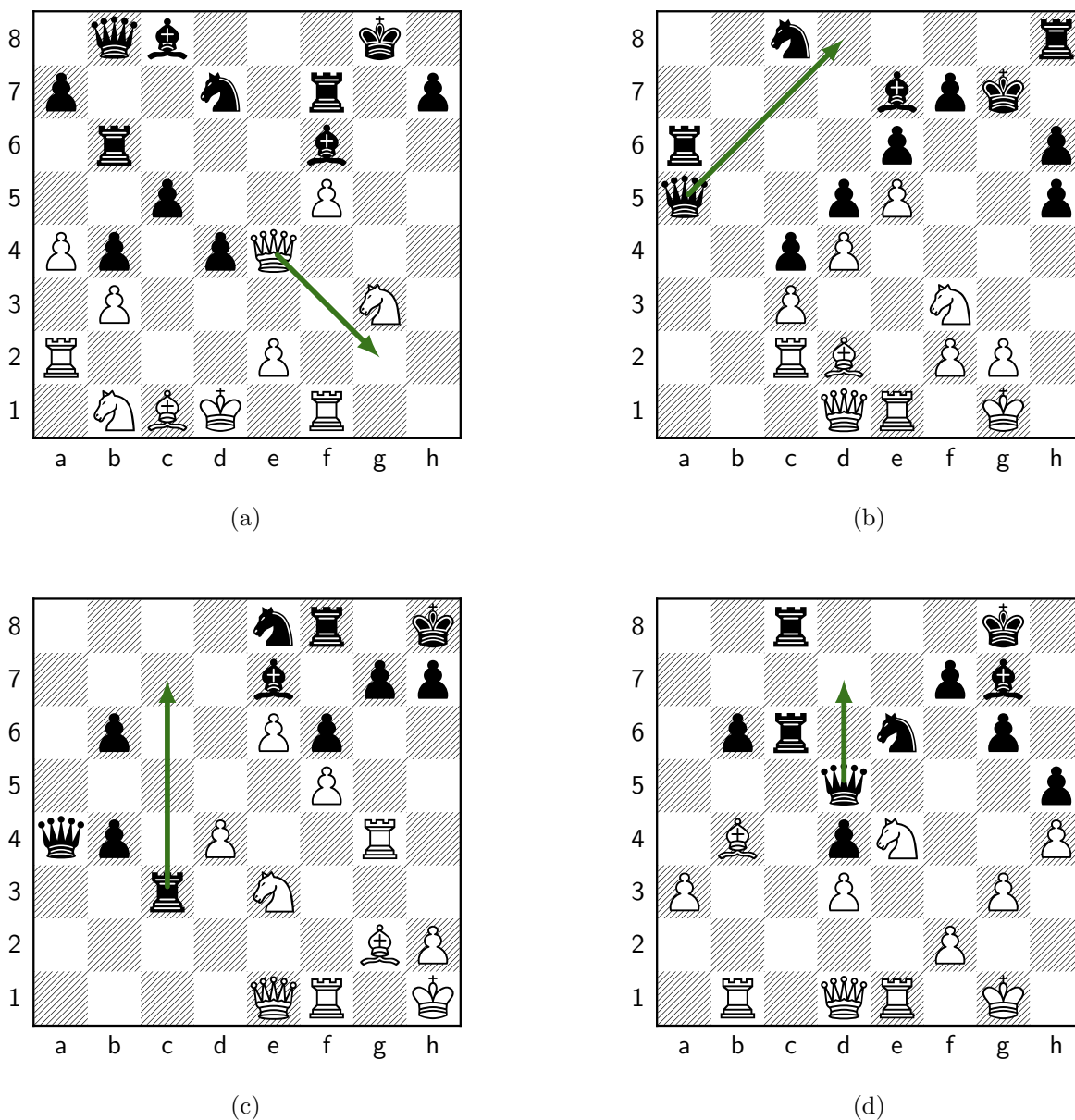
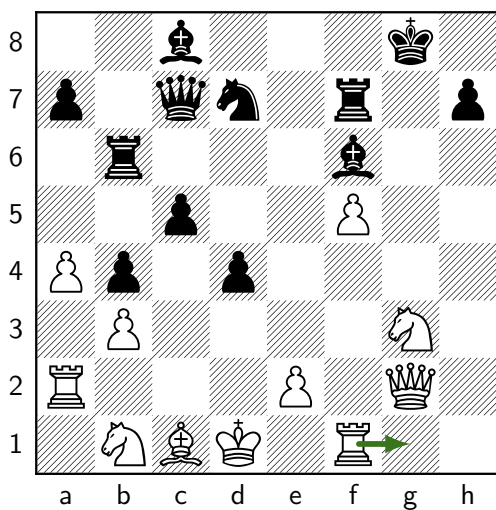
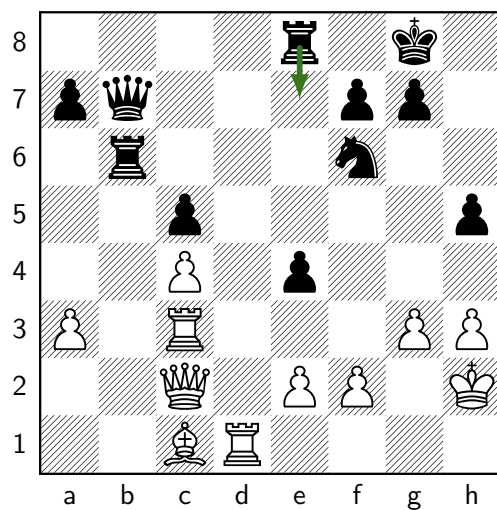


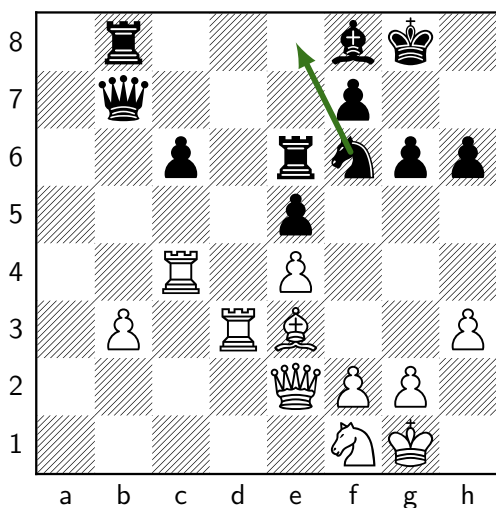
Figure 30: **Move 36, White: Qg2.** (a) Our position; Melanie plays Qg2 (e4g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2718` (ply 61) [structural distance  $d=100$ , QAP obj=21]: the master played Qd8; under  $\mu$  our Qe4 plays the role of that game’s Qa4; cross-type roles: our pa7 as their Nc8, our Pb3 as their Be7; transferred move e4g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2881` (ply 67) [structural distance  $d=112$ , QAP obj=14]: the master played Rc7; under  $\mu$  our Qe4 plays the role of that game’s Rc6 (a **cross-type** correspondence); cross-type roles: our Ra2 as their Ne8, our Pf5 as their Be7; transferred move e4g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#719` (ply 67) [structural distance  $d=110$ , QAP obj=12]: the master played Qd7; under  $\mu$  our Qe4 plays the role of that game’s Qd4; cross-type roles: our Pf5 as their Bg7, our Nb1 as their Pb6; transferred move e4g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qg2 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



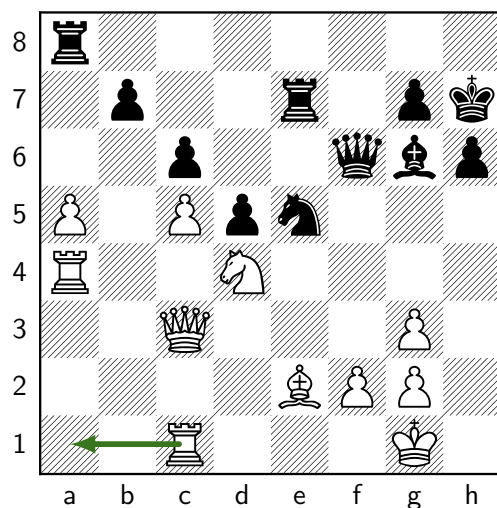
(a)



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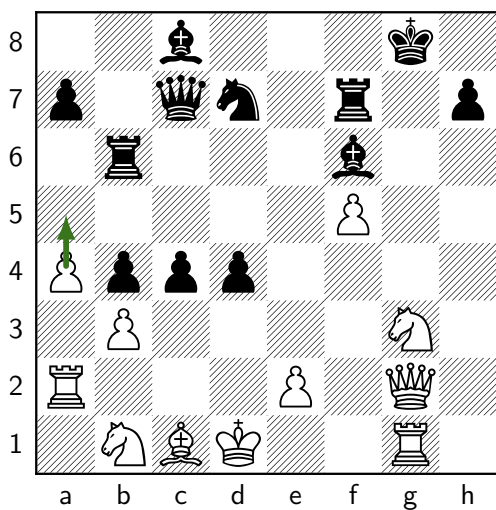


(c)

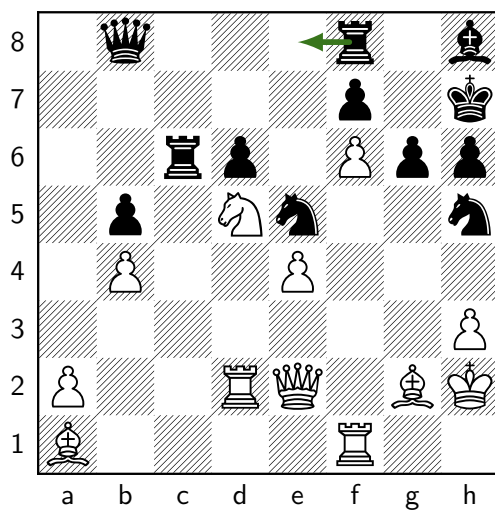


(d)

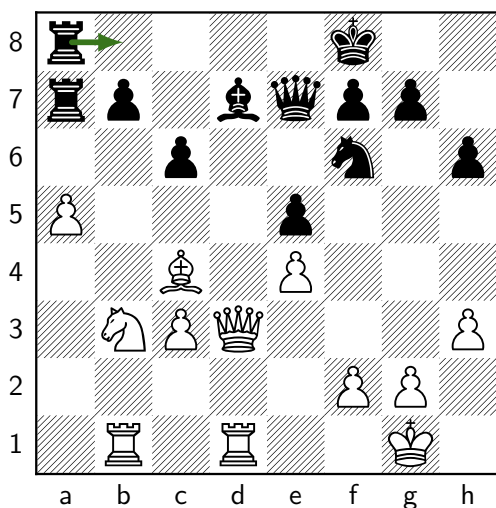
Figure 31: **Move 37, White: Rg1**. (a) Our position; Melanie plays Rg1 (f1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#1589 (ply 89) [structural distance  $d=104$ , QAP obj=20]: the master played Re7; under  $\mu$  our Rf1 plays the role of that game’s Re1; cross-type roles: our Nb1 as their Pa7, our Bc1 as their Pf7; transferred move f1g1 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#200 (ply 67) [structural distance  $d=103$ , QAP obj=19]: the master played Ne8; under  $\mu$  our Rf1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Rb8, our Ng3 as their Pc6; transferred move f1g1 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#56 (ply 68) [structural distance  $d=107$ , QAP obj=18]: the master played Rca1; under  $\mu$  our Rf1 plays the role of that game’s Rc1; cross-type roles: our rb6 as their Be2, our Pf5 as their Qc3; transferred move f1g1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rg1 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



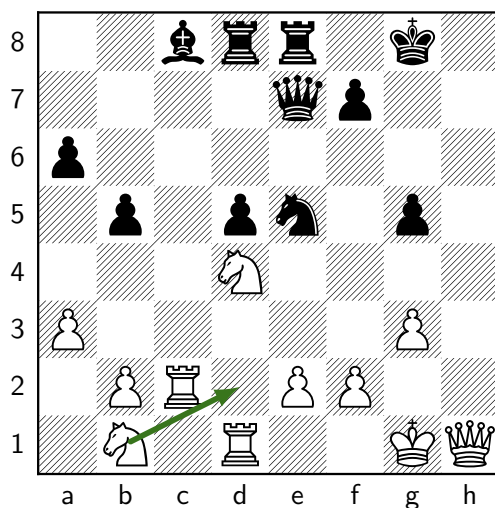
(a)



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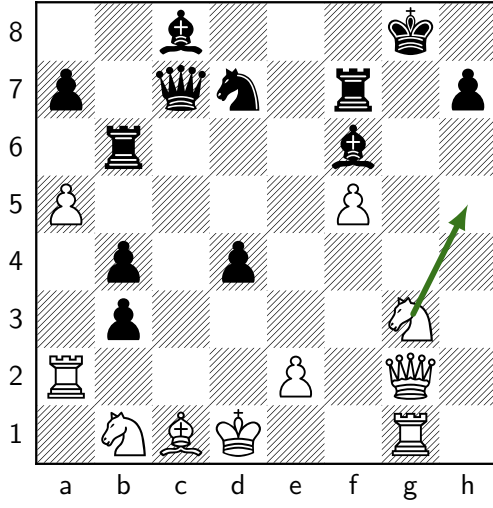


(c)

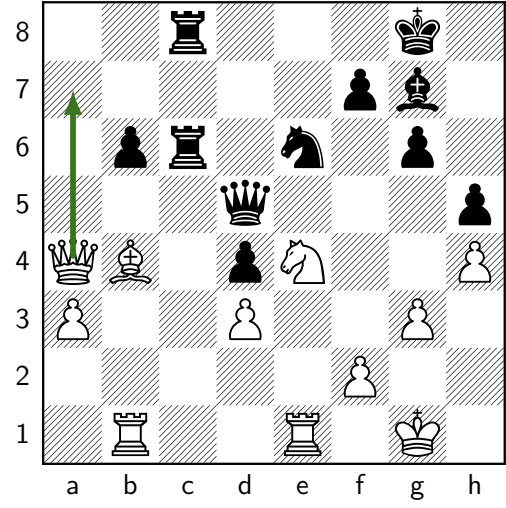


(d)

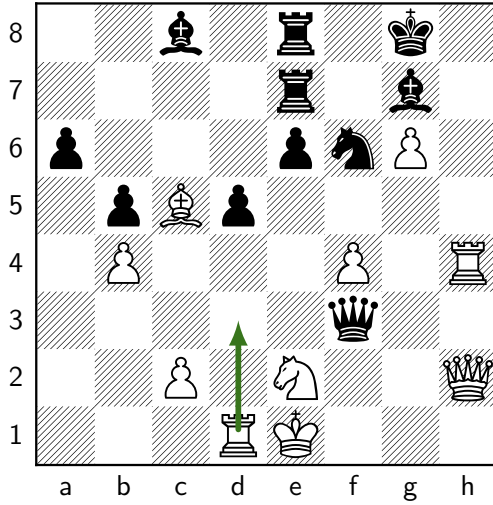
Figure 32: **Move 38, White: a5.** (a) Our position; Melanie plays a5 (a4a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2535` (ply 71) [structural distance  $d=89$ , QAP obj=47]: the master played  $R_e8$ ; under  $\mu$  our  $Pa4$  plays the role of that game’s  $Rf1$  (a **cross-type** correspondence); cross-type roles: our  $Ra2$  as their  $Qb8$ , our  $Pb3$  as their  $Bh8$ ; transferred move a4a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3651` (ply 59) [structural distance  $d=97$ , QAP obj=31]: the master played  $R_b8$ ; under  $\mu$  our  $Pa4$  plays the role of that game’s  $Ra1$  (a **cross-type** correspondence); cross-type roles: our  $Pb3$  as their  $Ra7$ , our  $Nb1$  as their  $Pb7$ ; transferred move a4a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1231` (ply 74) [structural distance  $d=93$ , QAP obj=24]: the master played  $N_d2$ ; under  $\mu$  our  $Pa4$  plays the role of that game’s  $Nb1$  (a **cross-type** correspondence); cross-type roles: our  $pd4$  as their  $Qh1$ , our  $pc4$  as their  $Rc2$ ; transferred move a4a5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a5 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



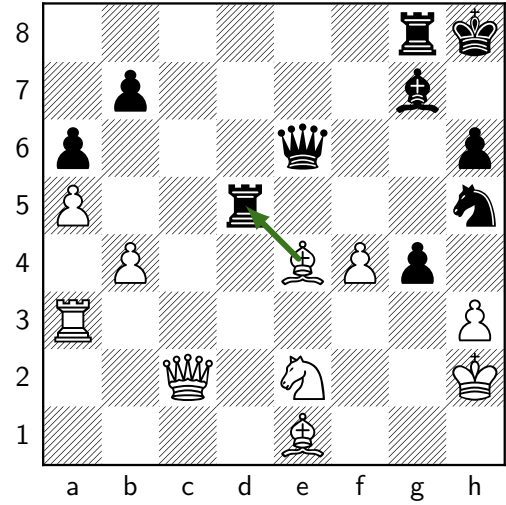
(a)



(b)

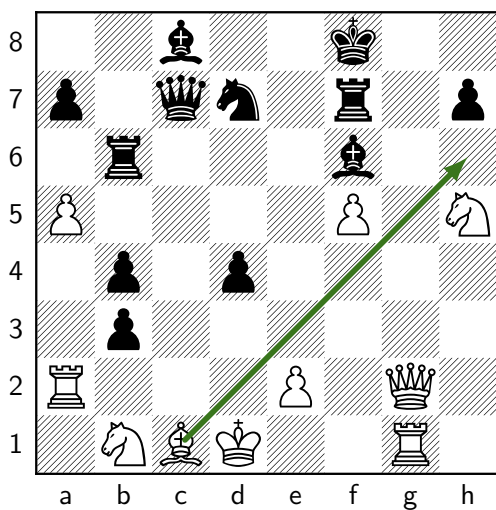


(c)

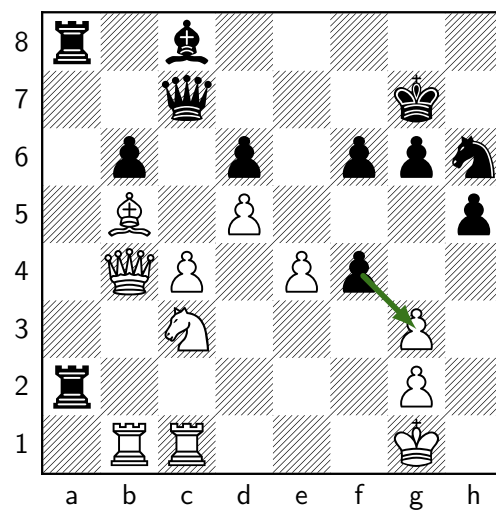


(d)

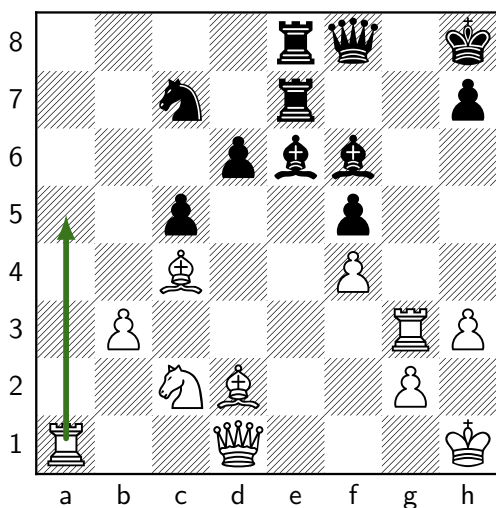
Figure 33: **Move 39, White: Nh5+ (g3h5).** (a) Our position; Melanie plays Nh5+ (g3h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#719` (ply 70) [structural distance  $d=113$ , QAP obj=24]: the master played Qa7; under  $\mu$  our Pa5 plays the role of that game’s Qa4 (a **cross-type** correspondence); cross-type roles: our Qg2 as their Rb1, our Nb1 as their Pf2; transferred move a5a6. (c) Retrieved from `ccr1_top_p0002#2357` (ply 70) [structural distance  $d=106$ , QAP obj=24]: the master played Rd3; under  $\mu$  our Rg1 plays the role of that game’s Rd1; cross-type roles: our rb6 as their Ne2, our Pa5 as their qf3; transferred move g1h1. (d) Retrieved from `ccr1_top_p0000#633` (ply 66) [structural distance  $d=106$ , QAP obj=23]: the master played Bxd5; under  $\mu$  our Pa5 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Be1, our Ra2 as their Qc2; transferred move a5b6. Synthesis: no retrieved analog transferred to Nh5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



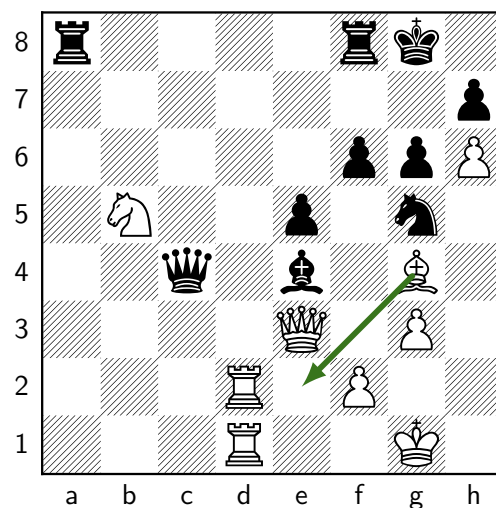
(a)



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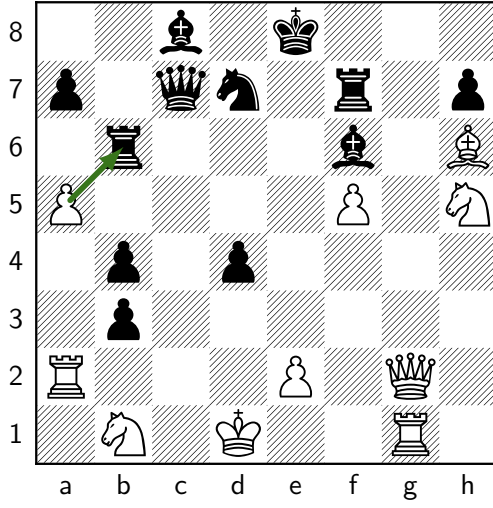


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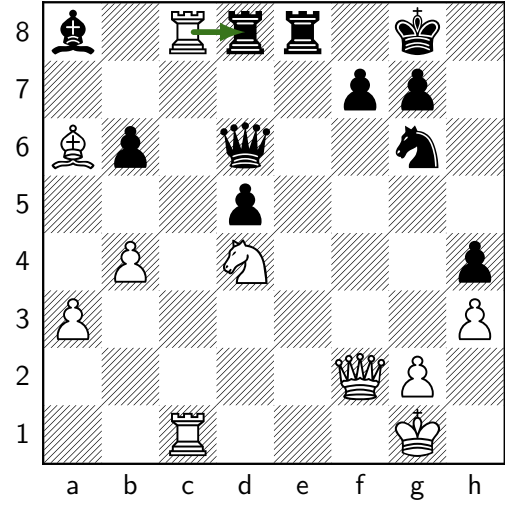


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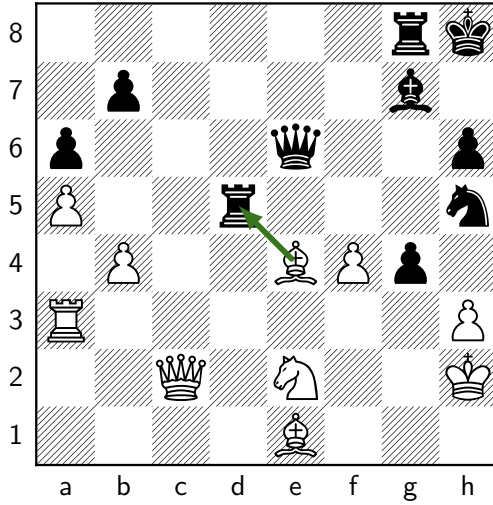
Figure 34: **Move 40, White: Bh6+.** (a) Our position; Melanie plays Bh6+ (c1h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#483` (ply 83) [structural distance  $d=131$ , QAP obj=13]: the master played `fxg3`; under  $\mu$  our `Bc1` plays the role of that game’s `Pf5` (a **cross-type** correspondence); cross-type roles: our `Pa5` as their `Ra8`, our `pd4` as their `Bc8`; transferred move `c1h6` (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#935` (ply 66) [structural distance  $d=136$ , QAP obj=10]: the master played `Ra5`; under  $\mu$  our `Bc1` plays the role of that game’s `Ra1` (a **cross-type** correspondence); cross-type roles: our `Rg1` as their `Qd1`, our `pb4` as their `Nc2`; transferred move `c1h6` (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3605` (ply 72) [structural distance  $d=123$ , QAP obj=28]: the master played `Be2`; under  $\mu$  our `Qg2` plays the role of that game’s `Bg4` (a **cross-type** correspondence); cross-type roles: our `Bc1` as their `Rd2`, our `Nh5` as their `Pf2`; transferred move `g2g5`. Synthesis: 2 retrieved analogs transferred to Bh6+ (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



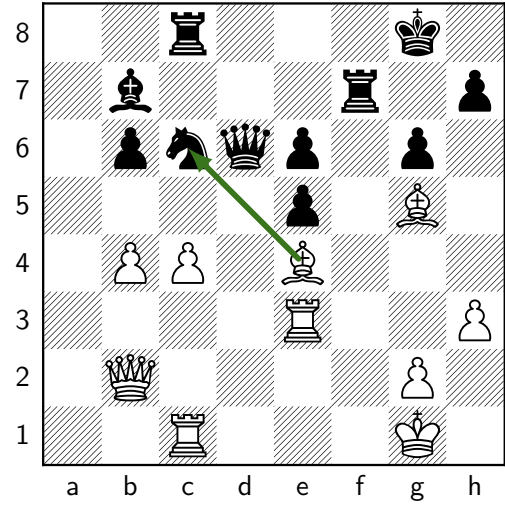
(a)



(b)

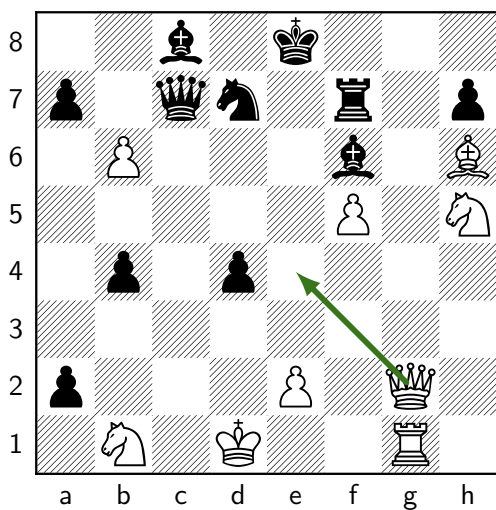


(c)

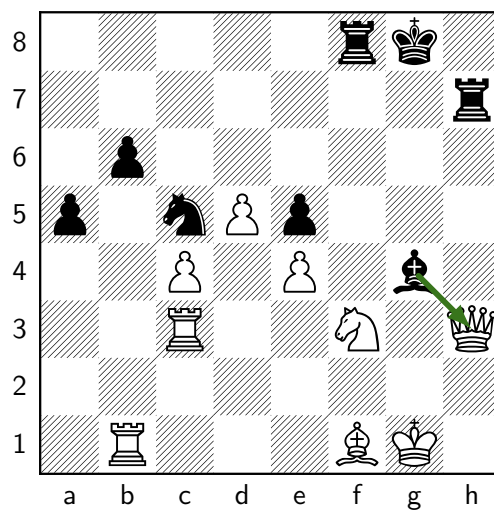


(d)

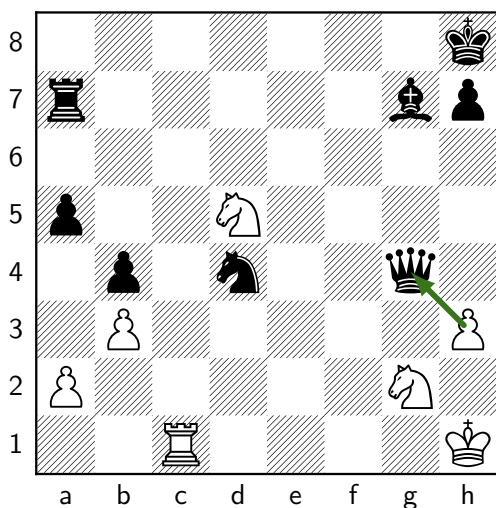
Figure 35: **Move 41, White: axb6.** (a) Our position; Melanie plays axb6 (a5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1109` (ply 78) [structural distance  $d=140$ , QAP obj=25]: the master played Rxd8; under  $\mu$  our *Pa5* plays the role of that game’s *Rc8* (a **cross-type** correspondence); cross-type roles: our *Bh6* as their *Rc1*, our *Rg1* as their *Qf2*; transferred move a5b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#633` (ply 66) [structural distance  $d=123$ , QAP obj=24]: the master played Bxd5; under  $\mu$  our *Pa5* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Ra2* as their *Qc2*, our *Rg1* as their *Ne2*; transferred move a5b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3759` (ply 70) [structural distance  $d=140$ , QAP obj=13]: the master played Bxc6; under  $\mu$  our *Pa5* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Nh5* as their *Qb2*, our *Qg2* as their *Pb4*; transferred move a5b6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to axb6 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



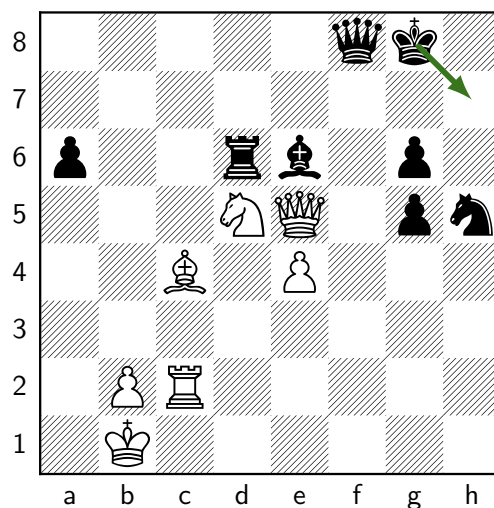
(a)



(b)



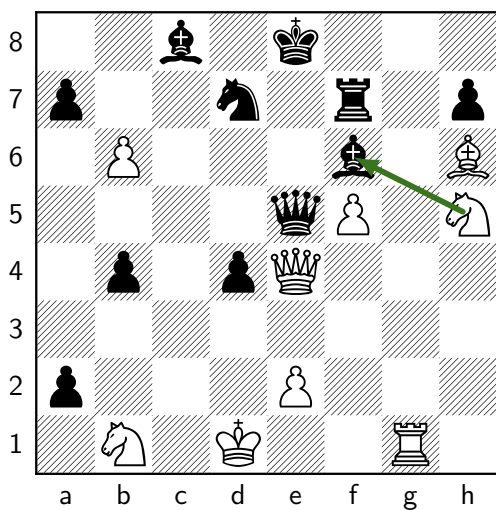
(c)



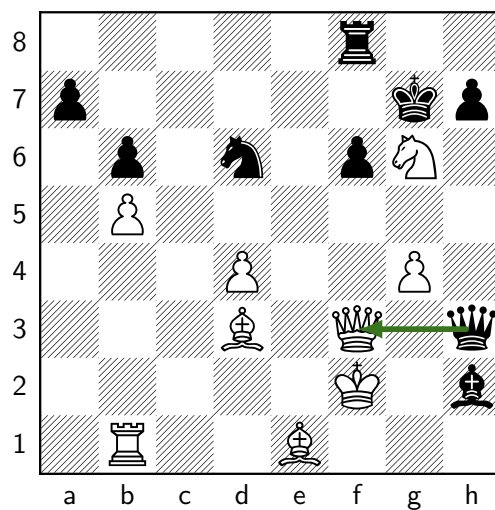
(d)

Figure 36: **Move 42, White: Qe4+.** (a) Our position; Melanie plays Qe4+ (g2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3176` (ply 79) [structural distance  $d=119$ , QAP obj=28]: the master played Bxh3; under  $\mu$  our Pb6 plays the role of that game's Bg5 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Rh7, our Nh5 as their Pa5; transferred move b6c7. (c) Retrieved from `ccr1_top_p0003#1246` (ply 76) [structural distance  $d=106$ , QAP obj=24]: the master played hxg4; under  $\mu$  our Pb6 plays the role of that game's Ph3; cross-type roles: our Qg2 as their Rc1, our Nh5 as their Pa2; transferred move b6c7. (d) Retrieved from `ccr1_top_p0002#1617` (ply 93) [structural distance  $d=118$ , QAP obj=21]: the master played Kh7; under  $\mu$  our Kd1 plays the role of that game's Kg1; cross-type roles: our Rg1 as their Qf8, our qc7 as their Rd6; transferred move d1e1. Synthesis: no retrieved analog transferred to Qe4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

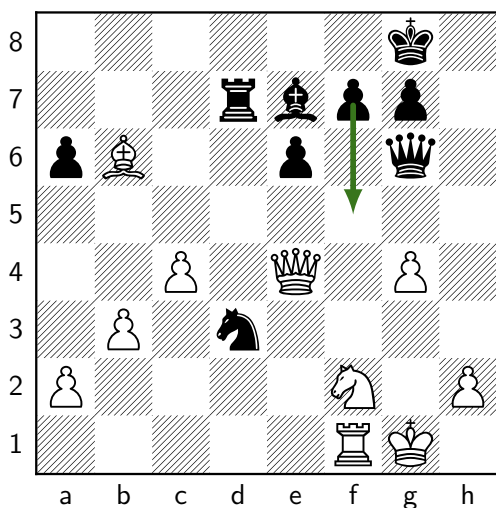




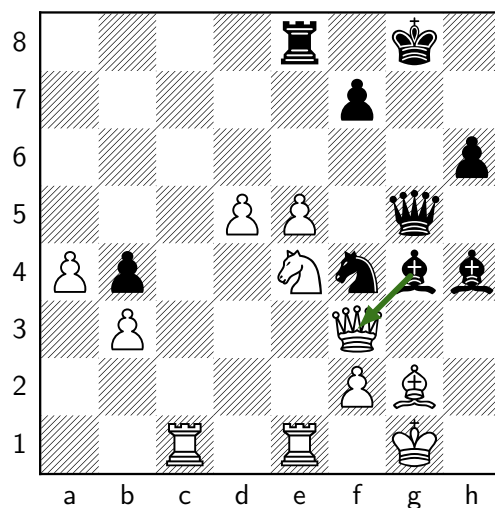
(a)



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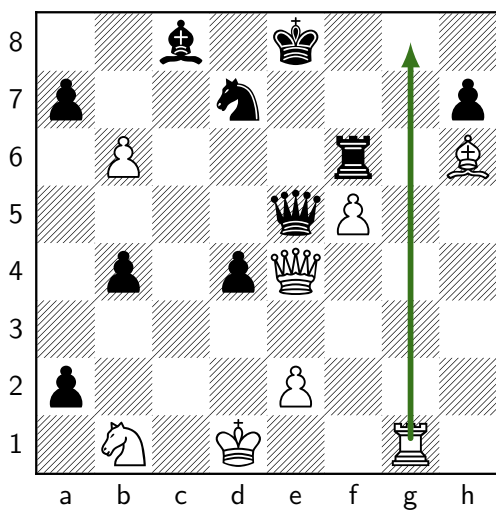


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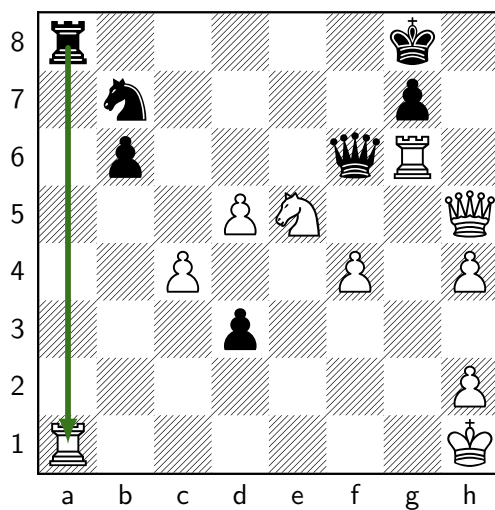


(d)

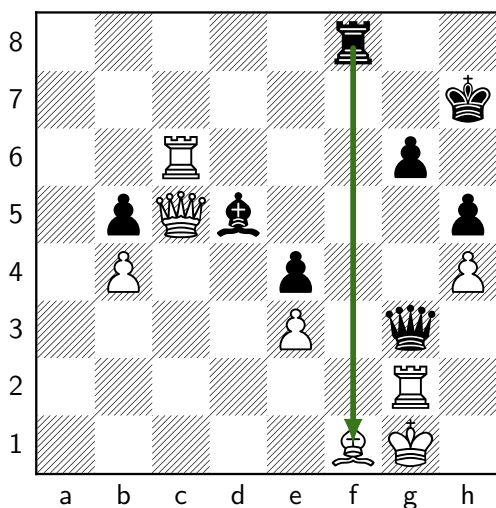
Figure 37: **Move 43, White: Nxf6+.** (a) Our position; Melanie plays Nxf6+ (h5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1526` (ply 87) [structural distance  $d=107$ , QAP obj=47]: the master played Qxf3+; under  $\mu$  our Qe4 plays the role of that game’s Qh6; cross-type roles: our Nh5 as their Rf8, our nd7 as their Pa7; transferred move e4e5. (c) Retrieved from `ccr1_top_p0001#3289` (ply 71) [structural distance  $d=111$ , QAP obj=43]: the master played f5; under  $\mu$  our Nh5 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Rd7, our Nb1 as their Be7; transferred move h5g3. (d) Retrieved from `ccr1_top_p0001#27` (ply 83) [structural distance  $d=117$ , QAP obj=40]: the master played Bxf3; under  $\mu$  our Qe4 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our bf6 as their pd5, our Nb1 as their Qg5; transferred move e4e5. Synthesis: no retrieved analog transferred to Nxf6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



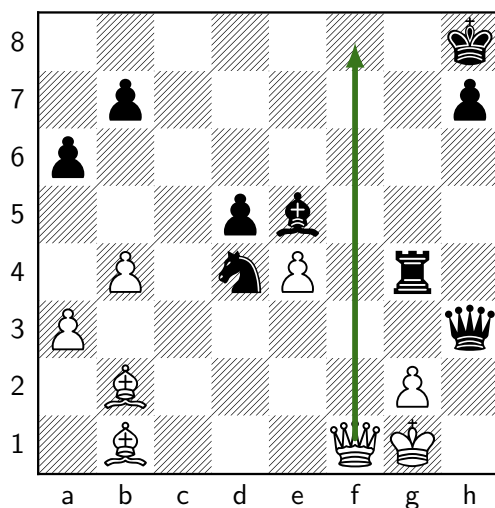
(a)



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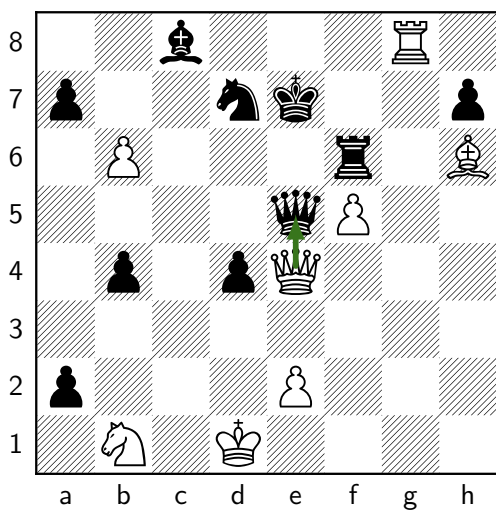


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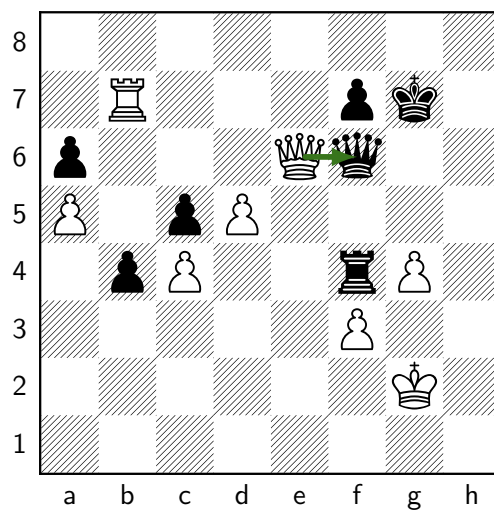


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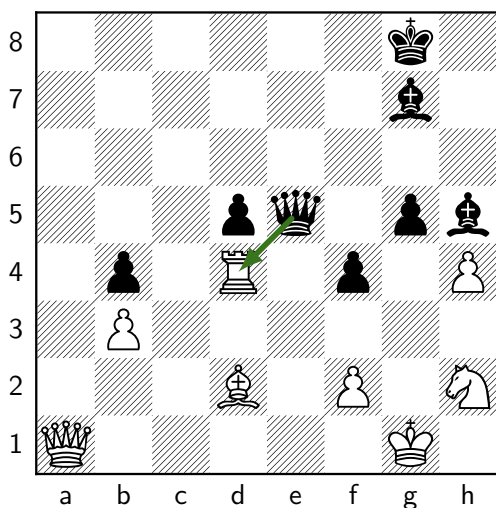
Figure 38: **Move 44, White: Rg8+.** (a) Our position; Melanie plays Rg8+ (g1g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#527` (ply 89) [structural distance  $d=101$ , QAP obj=61]: the master played Rxa1+; under  $\mu$  our Rg1 plays the role of that game’s Ra1; cross-type roles: our ph7 as their Nb7, our Nb1 as their Pb6; transferred move g1g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2613` (ply 71) [structural distance  $d=105$ , QAP obj=60]: the master played Rxf1+; under  $\mu$  our Rg1 plays the role of that game’s Rf1; cross-type roles: our Nb1 as their rc6, our rf6 as their qc5; transferred move g1g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3588` (ply 76) [structural distance  $d=93$ , QAP obj=33]: the master played Qf8+; under  $\mu$  our Rg1 plays the role of that game’s Qf1 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Bb1, our Pb6 as their Bb2; transferred move g1g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rg8+ (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



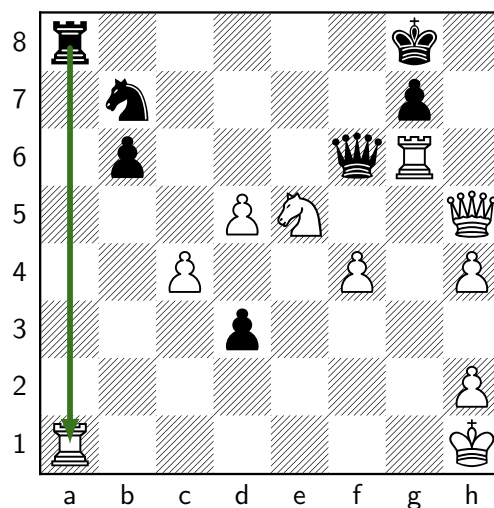
(a)



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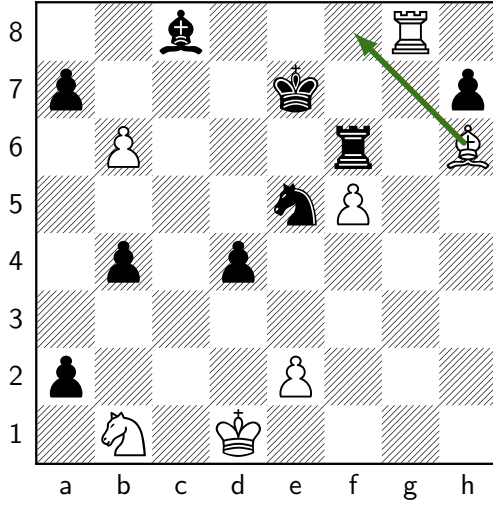


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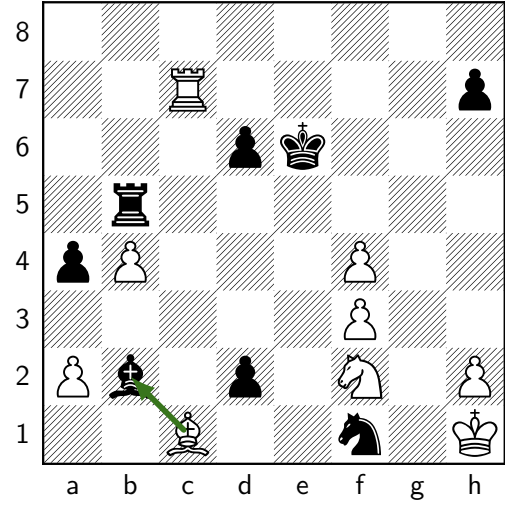


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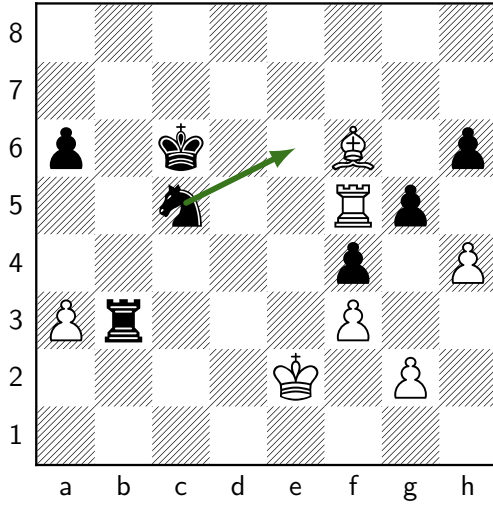
Figure 39: **Move 45, White: Qxe5+.** (a) Our position; Melanie plays Qxe5+ (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#453` (ply 88) [structural distance  $d=120$ , QAP obj=60]: the master played Qxf6+; under  $\mu$  our Qe4 plays the role of that game's Qe6; cross-type roles: our Rg8 as their pb4, our Nb1 as their Pc4; transferred move e4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#363` (ply 87) [structural distance  $d=113$ , QAP obj=33]: the master played Qxd4; under  $\mu$  our Qe4 plays the role of that game's Qe4; cross-type roles: our Pe2 as their Bg7, our pa7 as their Bh5; transferred move e4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#527` (ply 89) [structural distance  $d=116$ , QAP obj=61]: the master played Rxa1+; under  $\mu$  our Pb6 plays the role of that game's Ra1 (a **cross-type** correspondence); cross-type roles: our Bh6 as their Nb7, our Nb1 as their Pb6; transferred move b6a7. Synthesis: 2 retrieved analogs transferred to Qxe5+ (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



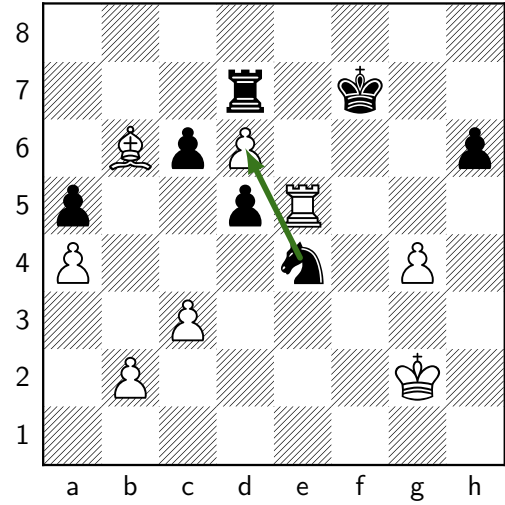
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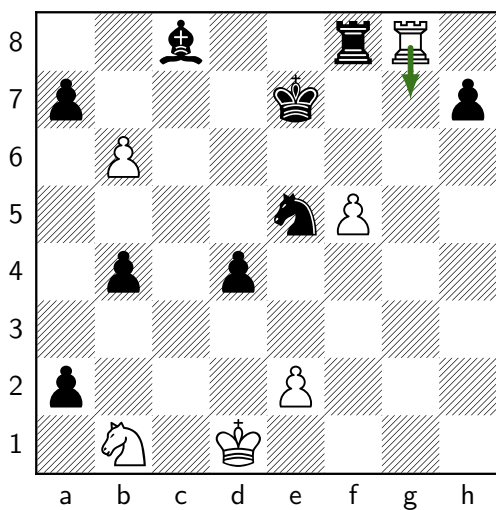


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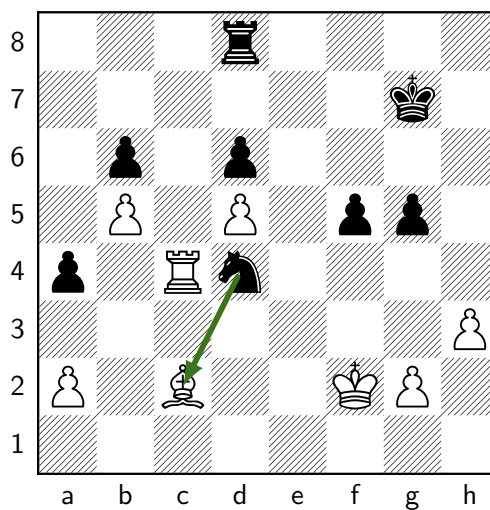


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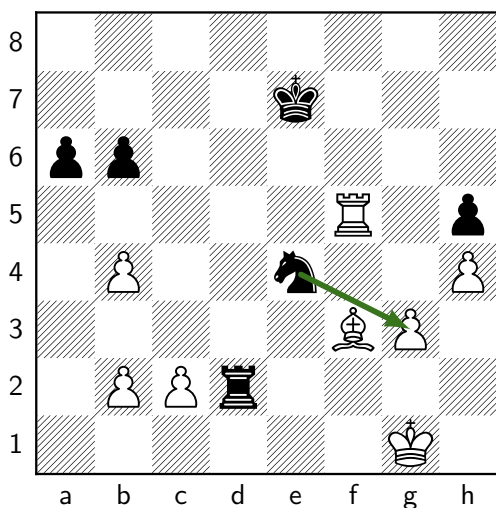
Figure 40: **Move 46, White: Bf8+.** (a) Our position; Melanie plays Bf8+ (h6f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1675` (ply 90) [structural distance  $d=56$ , QAP obj=40]: the master played Bxb2; under  $\mu$  our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bh6 as their nf1, our pa2 as their bb2; transferred move b1a3. (c) Retrieved from `ccr1_top_p0003#589` (ply 105) [structural distance  $d=54$ , QAP obj=36]: the master played Ne6; under  $\mu$  our Bh6 plays the role of that game’s Nc4 (a **cross-type** correspondence); cross-type roles: our Pb6 as their bf6, our Nb1 as their pa3; transferred move h6e3. (d) Retrieved from `ccr1_top_p0002#317` (ply 91) [structural distance  $d=54$ , QAP obj=34]: the master played Nxd6; under  $\mu$  our Bh6 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Rd7, our pa2 as their bb6; transferred move h6g5. Synthesis: no retrieved analog transferred to Bf8+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



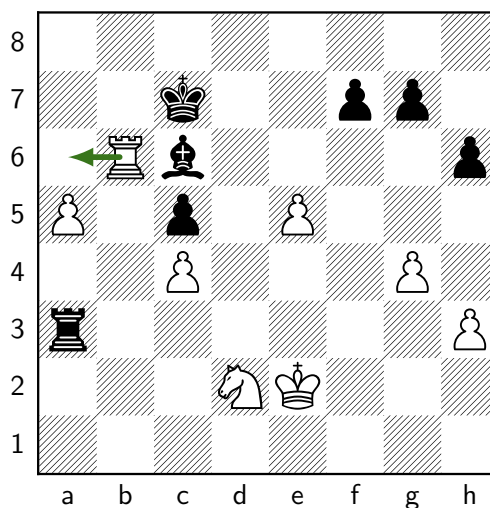
(a)



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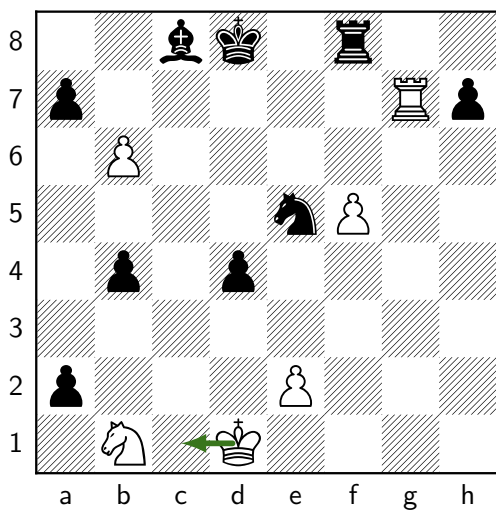


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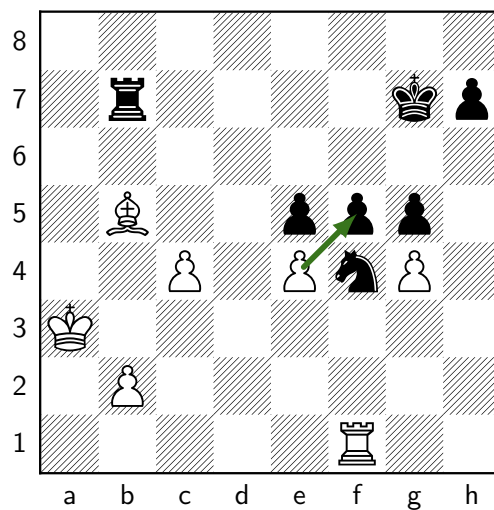


(d)

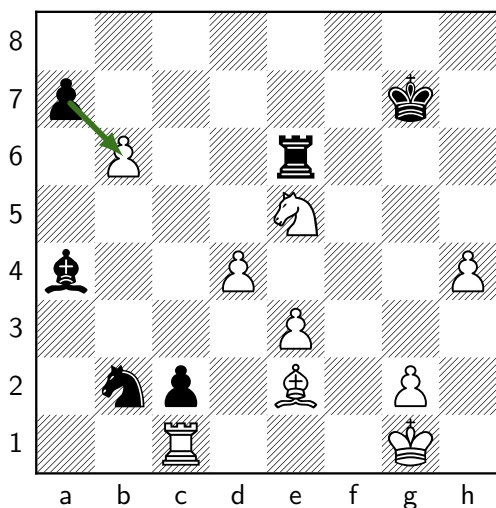
Figure 41: **Move 47, White: Rg7+.** (a) Our position; Melanie plays Rg7+ (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#691` (ply 81) [structural distance  $d=46$ , QAP obj=19]: the master played Nxc2; under  $\mu$  our Rg8 plays the role of that game’s Nd5 (a **cross-type** correspondence); cross-type roles: our ne5 as their Rd8, our Nb1 as their Pb6; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#105` (ply 83) [structural distance  $d=42$ , QAP obj=16]: the master played Nxg3; under  $\mu$  our Rg8 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Pb6, our bc8 as their rf5; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3687` (ply 100) [structural distance  $d=47$ , QAP obj=15]: the master played Ra6; under  $\mu$  our Rg8 plays the role of that game’s Rb6; cross-type roles: our Pe2 as their Nd2, our pd4 as their ra3; transferred move g8g7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rg7+ (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



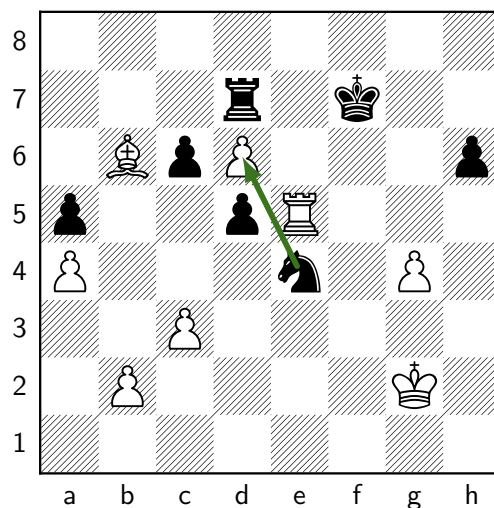
(a)



(b)

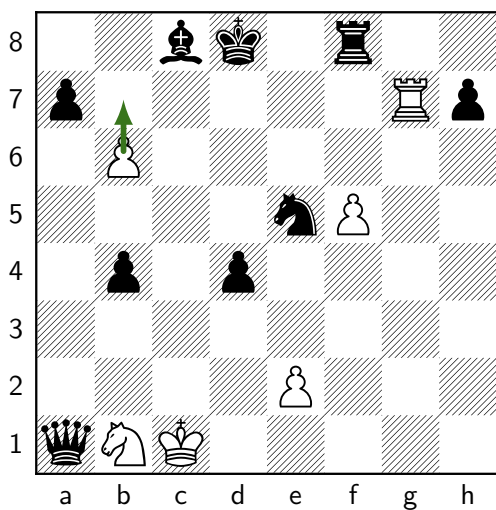


(c)

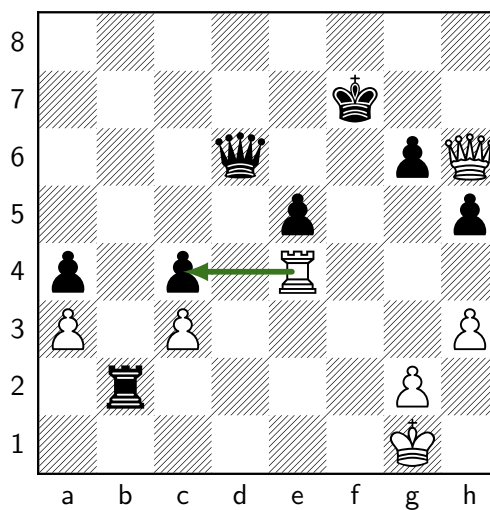


(d)

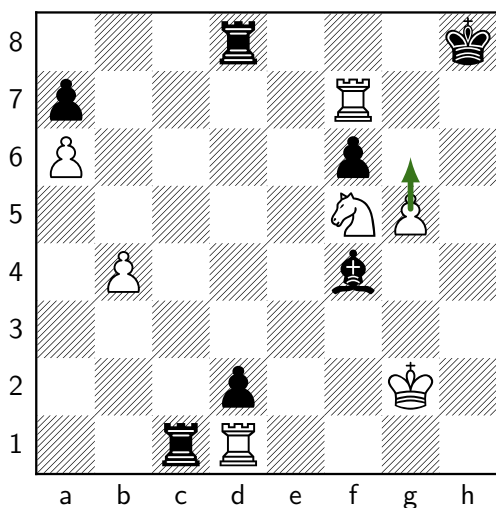
Figure 42: **Move 48, White: Kc1.** (a) Our position; Melanie plays Kc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1639` (ply 98) [structural distance  $d=49$ , QAP obj=37]: the master played `exf5`; under  $\mu$  our *Pb6* plays the role of that game’s *Pe4*; cross-type roles: our *ph7* as their *nf4*, our *Nb1* as their *Bb5*; transferred move `b6a7`. (c) Retrieved from `ccr1_top_p0001#293` (ply 85) [structural distance  $d=49$ , QAP obj=33]: the master played `axb6`; under  $\mu$  our *Pb6* plays the role of that game’s *Pa2*; cross-type roles: our *pa2* as their *Re6*, our *pd4* as their *Ba4*; transferred move `b6a7`. (d) Retrieved from `ccr1_top_p0002#317` (ply 91) [structural distance  $d=51$ , QAP obj=33]: the master played `Nxd6`; under  $\mu$  our *Nb1* plays the role of that game’s *Ne5*; cross-type roles: our *Pb6* as their *Rd7*, our *Rg7* as their *bb6*; transferred move `b1a3`. Synthesis: no retrieved analog transferred to Kc1 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



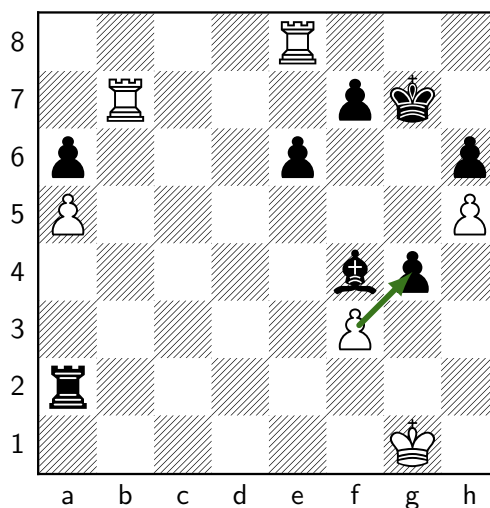
(a)



(b)



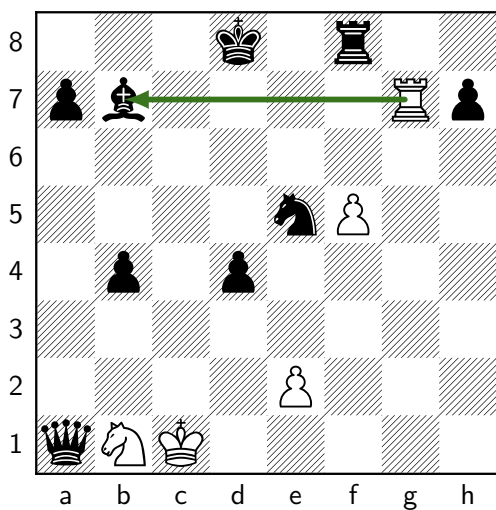
(c)



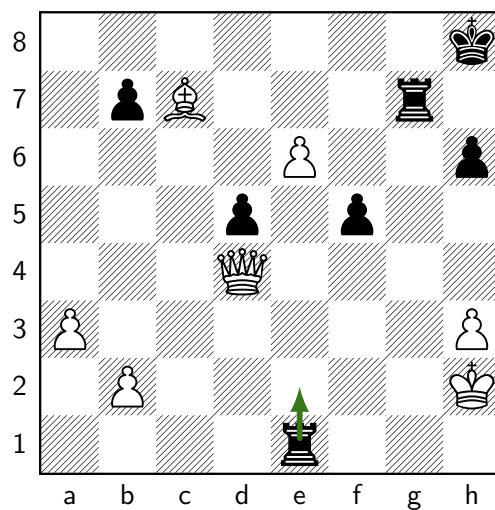
(d)

Figure 43: **Move 49, White: b7**. (a) Our position; Melanie plays b7 (b6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2645` (ply 90) [structural distance  $d=58$ , QAP obj=23]: the master played  $Rxc4$ ; under  $\mu$  our  $Pb6$  plays the role of that game’s  $Re4$  (a **cross-type** correspondence); cross-type roles: our  $ne5$  as their  $Pa3$ , our  $Nb1$  as their  $ph5$ ; transferred move **b6b7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#1971` (ply 94) [structural distance  $d=60$ , QAP obj=16]: the master played  $g6$ ; under  $\mu$  our  $Pb6$  plays the role of that game’s  $Pg5$ ; cross-type roles: our  $Nb1$  as their  $rc1$ , our  $qa1$  as their  $Rd1$ ; transferred move **b6b7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#1044` (ply 96) [structural distance  $d=54$ , QAP obj=14]: the master played  $fxg4$ ; under  $\mu$  our  $Pb6$  plays the role of that game’s  $Pf3$ ; cross-type roles: our  $qa1$  as their  $ra2$ , our  $Nb1$  as their  $Pa5$ ; transferred move **b6b7 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to b7 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

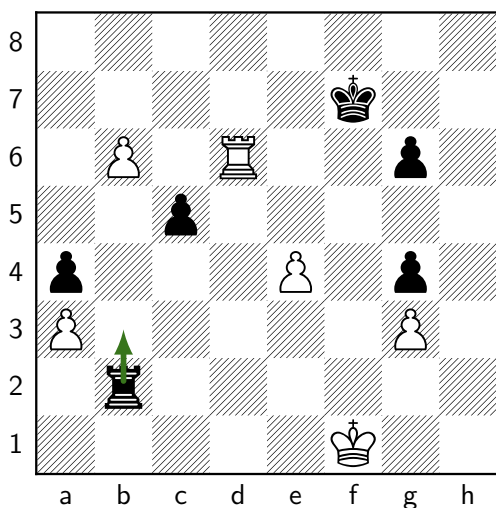




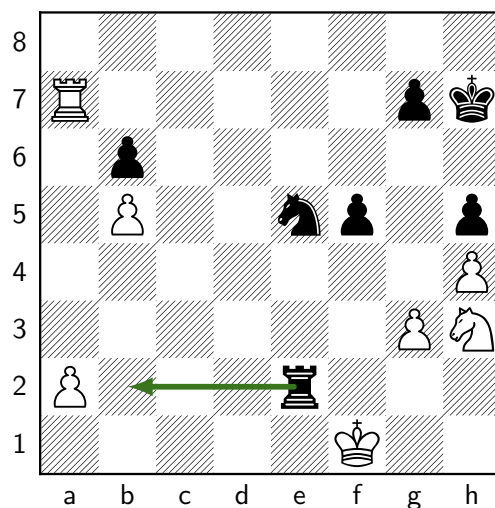
(a)



(b)

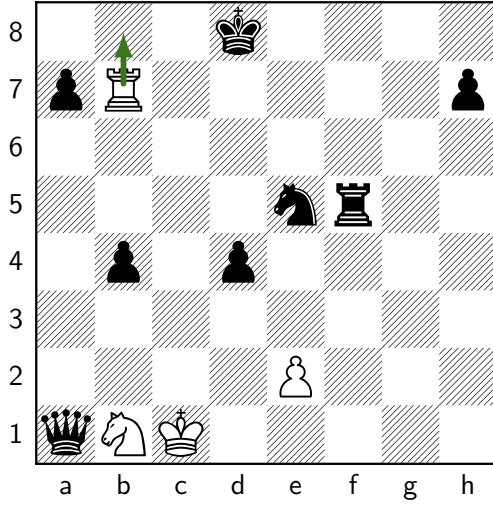


(c)

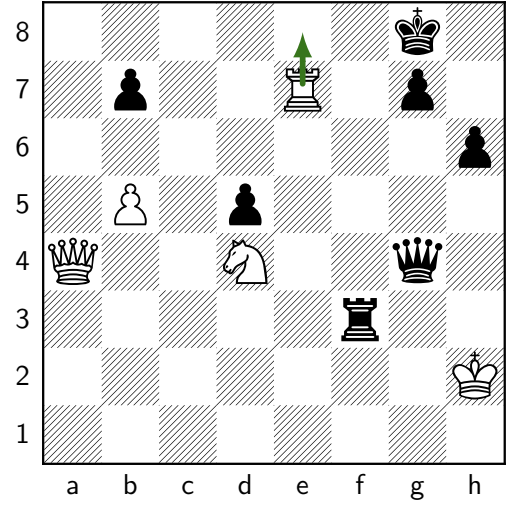


(d)

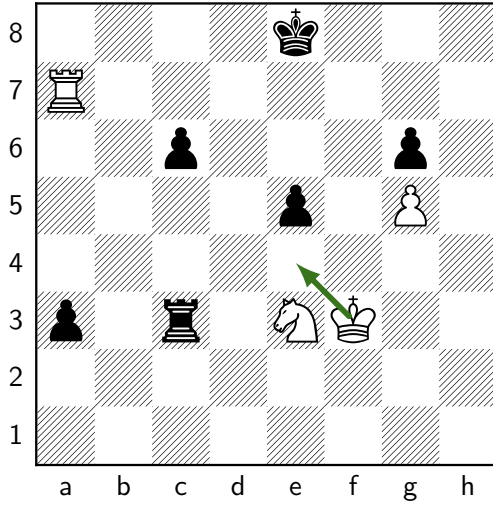
Figure 44: **Move 50, White: Rxb7**. (a) Our position; Melanie plays Rxb7 (g7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3673` (ply 103) [structural distance  $d=43$ , QAP obj=29]: the master played Re2+; under  $\mu$  our Rg7 plays the role of that game’s Re8; cross-type roles: our rf8 as their bc7, our Nb1 as their Rg7; transferred move g7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1746` (ply 109) [structural distance  $d=47$ , QAP obj=4]: the master played Rb3; under  $\mu$  our Rg7 plays the role of that game’s Rb7; cross-type roles: our qa1 as their rd6, our Nb1 as their Pg6; transferred move g7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3591` (ply 83) [structural distance  $d=40$ , QAP obj=3]: the master played Rb2; under  $\mu$  our Rg7 plays the role of that game’s Re7; cross-type roles: our qa1 as their ra7, our Nb1 as their Pg7; transferred move g7b7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rxb7 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



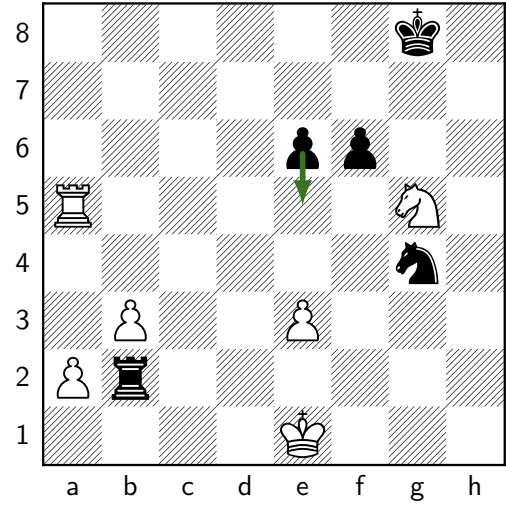
(a)



(b)

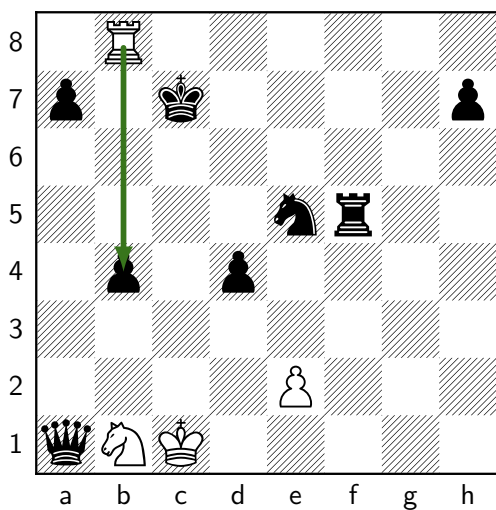


(c)

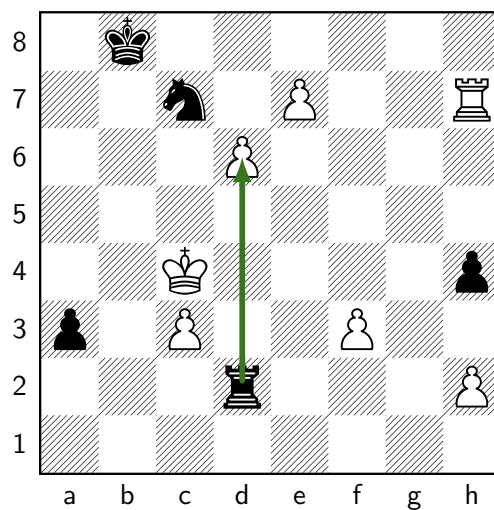


(d)

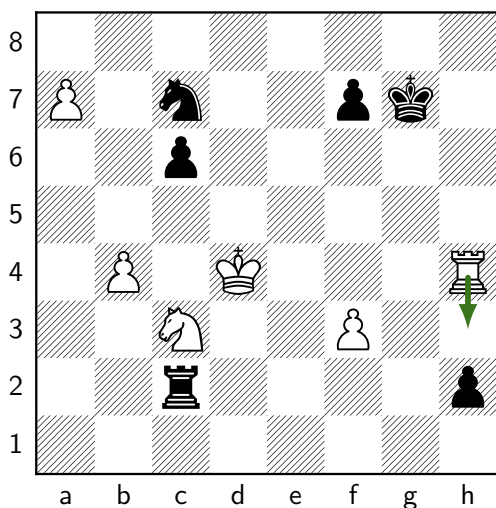
Figure 45: **Move 51, White: Rb8+.** (a) Our position; Melanie plays Rb8+ (b7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#784` (ply 86) [structural distance  $d=43$ , QAP obj=17]: the master played Re8+; under  $\mu$  our Rb7 plays the role of that game’s Re7; cross-type roles: our Nb1 as their rf3, our rf5 as their Qa4; transferred move b7b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2821` (ply 86) [structural distance  $d=42$ , QAP obj=29]: the master played Ke4; under  $\mu$  our Kc1 plays the role of that game’s Kf3; cross-type roles: our qa1 as their rc3, our ne5 as their Pg5; transferred move c1d1. (d) Retrieved from `ccr1_top_p0002#948` (ply 117) [structural distance  $d=44$ , QAP obj=27]: the master played e5; under  $\mu$  our Pe2 plays the role of that game’s Pe3; cross-type roles: our qa1 as their Pf6, our ne5 as their ra5; transferred move e2e4. Synthesis: 1 retrieved analog transferred to Rb8+ (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



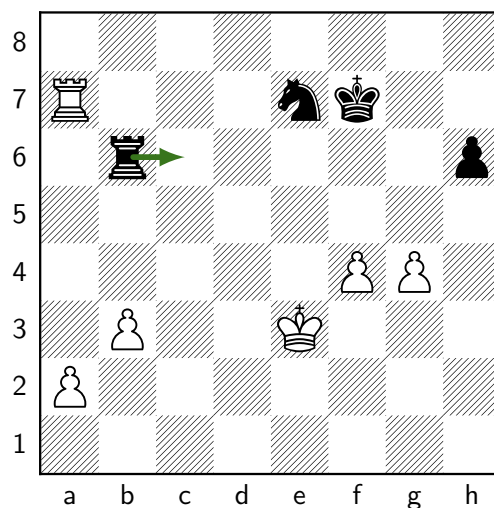
(a)



(b)

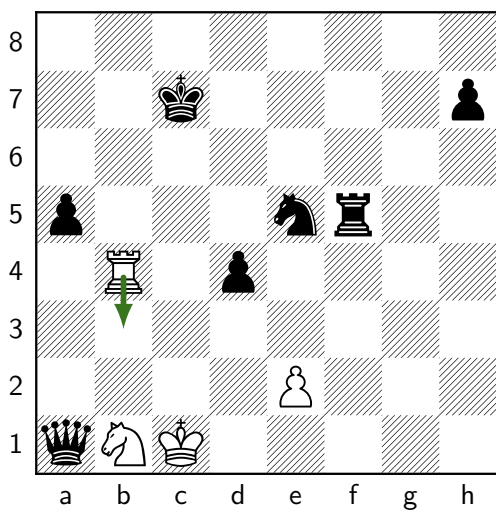


(c)

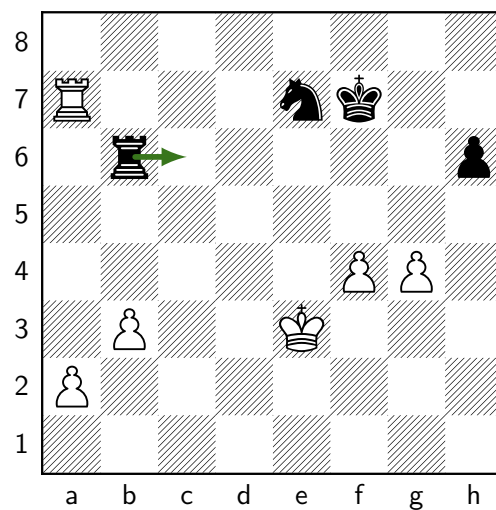


(d)

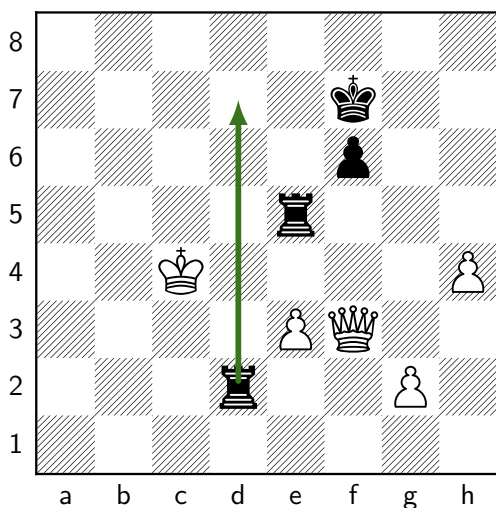
Figure 46: **Move 52, White: Rxb4**. (a) Our position; Melanie plays Rxb4 (b8b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2514` (ply 111) [structural distance  $d=38$ , QAP obj=14]: the master played Rxd6; under  $\mu$  our Rb8 plays the role of that game’s Rd7; cross-type roles: our ph7 as their rh7, our qa1 as their pd6; transferred move b8b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2859` (ply 94) [structural distance  $d=34$ , QAP obj=14]: the master played Rh3; under  $\mu$  our Rb8 plays the role of that game’s Rh4; cross-type roles: our qa1 as their rc2, our ne5 as their Pf3; transferred move b8b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1999` (ply 115) [structural distance  $d=38$ , QAP obj=11]: the master played Rc6; under  $\mu$  our Rb8 plays the role of that game’s Rb3; cross-type roles: our qa1 as their ra7, our ne5 as their pf4; transferred move b8b4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxb4 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



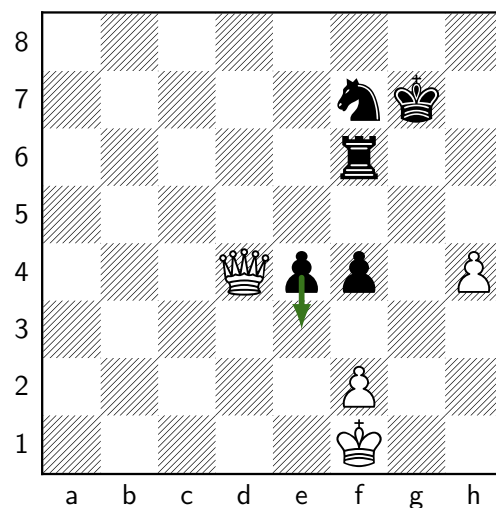
(a)



(b)



(c)



(d)

Figure 47: **Move 53, White: Rb3.** (a) Our position; Melanie plays Rb3 (b4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1999` (ply 115) [structural distance  $d=36$ , QAP obj=10]: the master played Rc6; under  $\mu$  our Rb4 plays the role of that game’s Rb3; cross-type roles: our qal as their ra7, our ne5 as their pg4; transferred move b4b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1754` (ply 107) [structural distance  $d=38$ , QAP obj=6]: the master played Rd7; under  $\mu$  our Rb4 plays the role of that game’s Rd7; cross-type roles: our Nb1 as their Pf6, our ne5 as their Re5; transferred move b4b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1372` (ply 111) [structural distance  $d=38$ , QAP obj=31]: the master played e3; under  $\mu$  our Pe2 plays the role of that game’s Pe5; cross-type roles: our Rb4 as their Nf7, our Nb1 as their Rf6; transferred move e2e3. Synthesis: 2 retrieved analogs transferred to Rb3 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

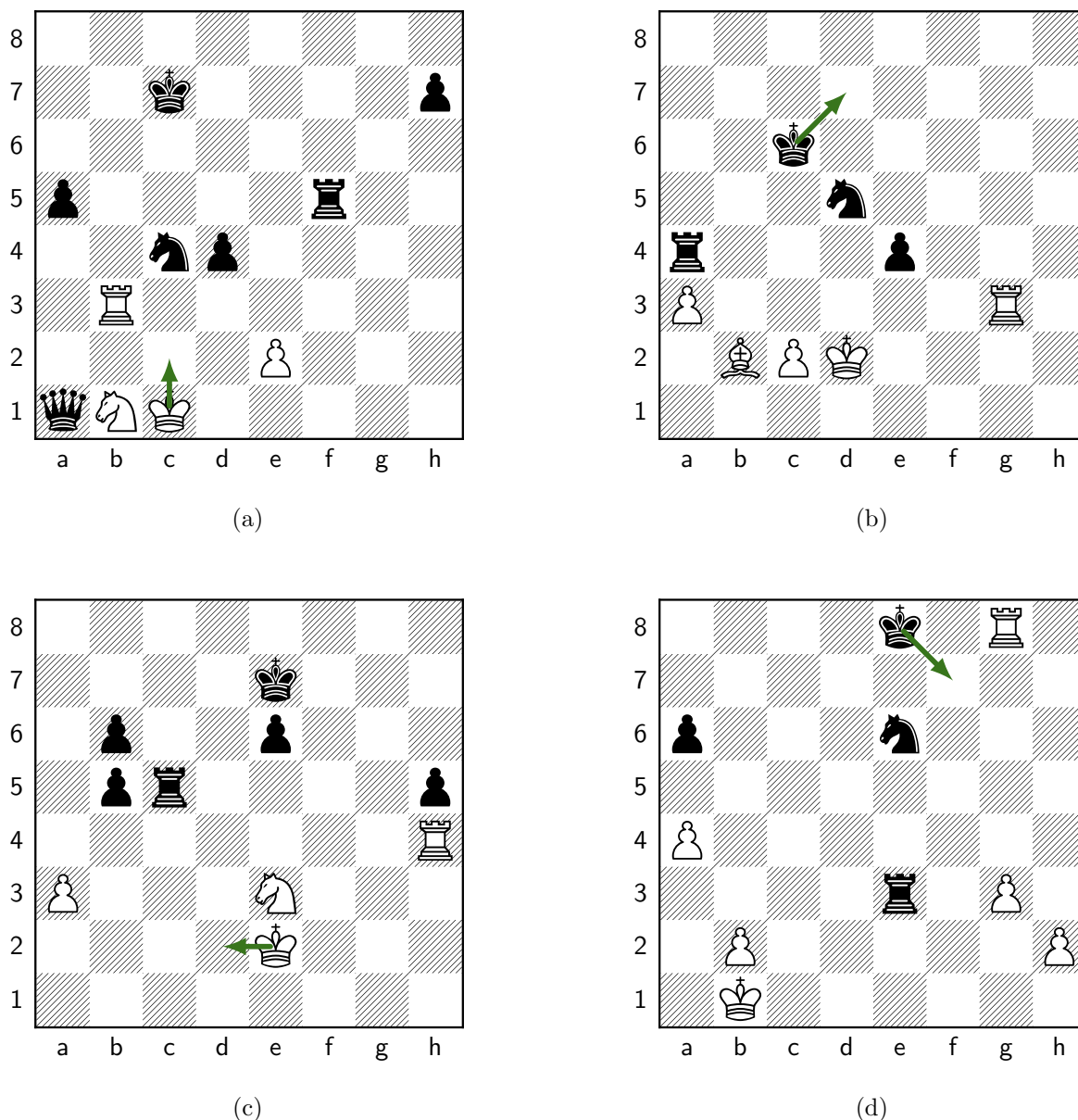
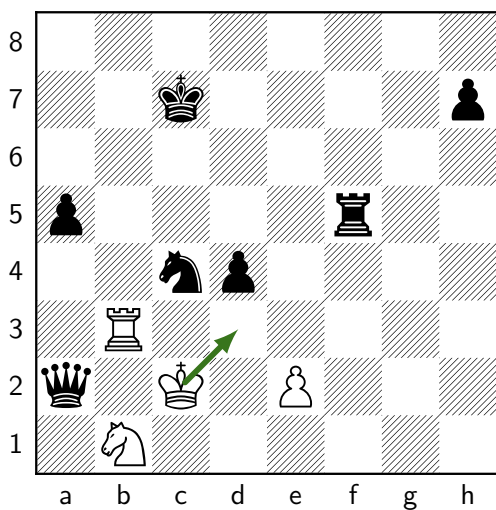
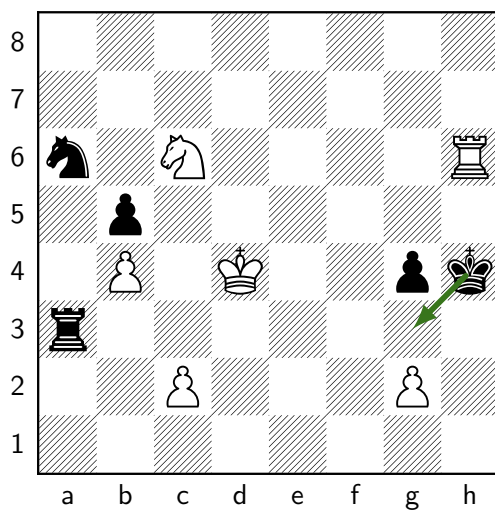


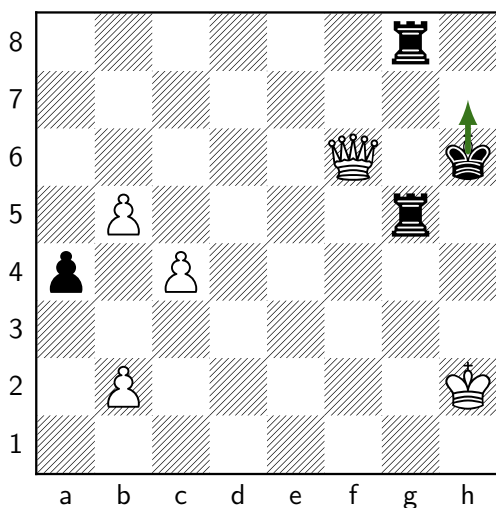
Figure 48: **Move 54, White: Kc2.** (a) Our position; Melanie plays Kc2 (c1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1666` (ply 119) [structural distance  $d=48$ , QAP obj=9]: the master played Kd7; under  $\mu$  our Kc1 plays the role of that game’s Kc3; cross-type roles: our qa1 as their Ra4, our Nb1 as their pa3; transferred move c1c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2567` (ply 110) [structural distance  $d=48$ , QAP obj=8]: the master played Kd2; under  $\mu$  our Kc1 plays the role of that game’s Ke2; cross-type roles: our qa1 as their Rh4, our rf5 as their pb5; transferred move c1c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1898` (ply 93) [structural distance  $d=44$ , QAP obj=2]: the master played Kf7; under  $\mu$  our Kc1 plays the role of that game’s Ke1; cross-type roles: our nc4 as their rg8, our rf5 as their pa4; transferred move c1c2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kc2 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



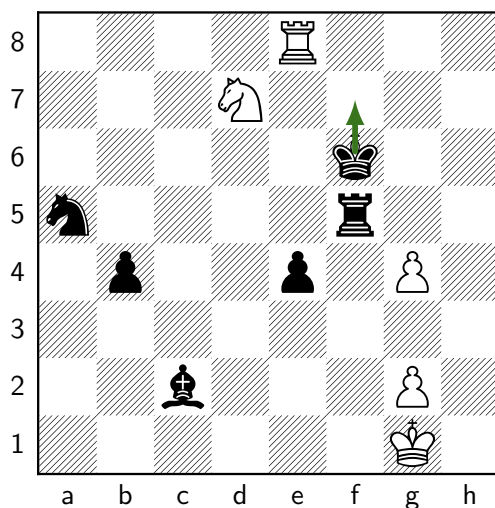
(a)



(b)

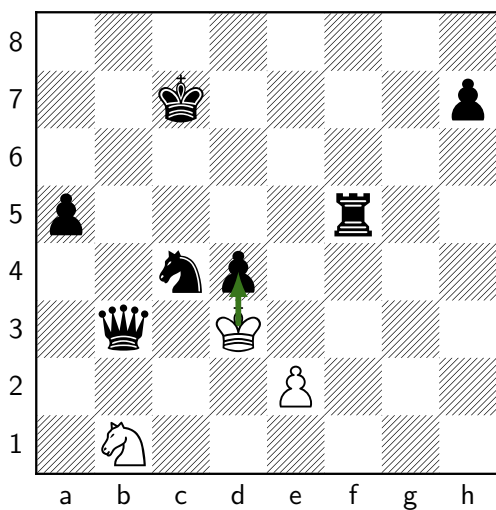


(c)

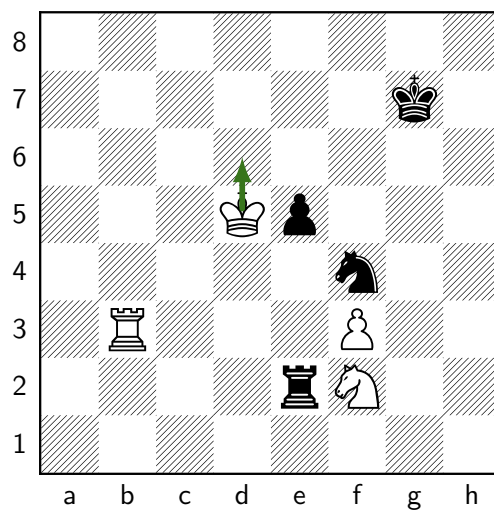


(d)

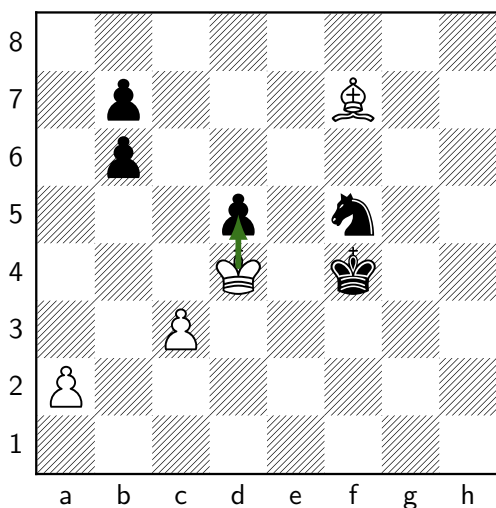
Figure 49: **Move 55, White: Kd3.** (a) Our position; Melanie plays Kd3 (c2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#310` (ply 97) [structural distance  $d=46$ , QAP obj=68]: the master played Kg3; under  $\mu$  our Kc2 plays the role of that game’s Kh5; cross-type roles: our qa2 as their rh6, our rf5 as their Pg4; transferred move c2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1092` (ply 111) [structural distance  $d=48$ , QAP obj=73]: the master played Kh7; under  $\mu$  our Kc2 plays the role of that game’s Kh3; cross-type roles: our Nb1 as their Rg8, our nc4 as their Pa4; transferred move c2d1. (d) Retrieved from `ccr1_top_p0002#2242` (ply 113) [structural distance  $d=49$ , QAP obj=69]: the master played Kf7; under  $\mu$  our Kc2 plays the role of that game’s Kf3; cross-type roles: our nc4 as their re8, our qa2 as their nd7; transferred move c2c1. Synthesis: 1 retrieved analog transferred to Kd3 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



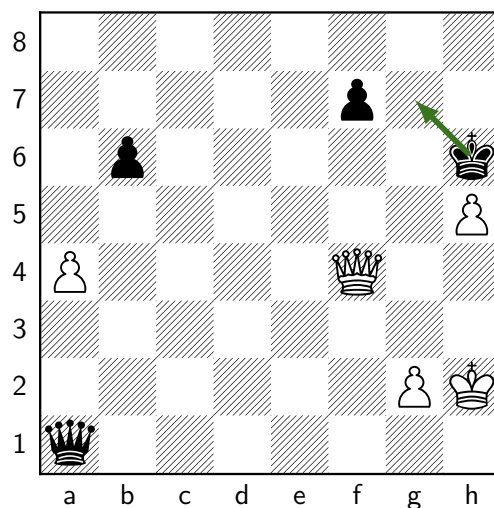
(a)



(b)



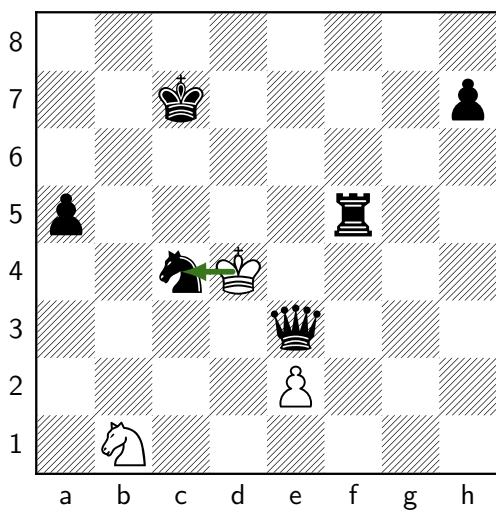
(c)



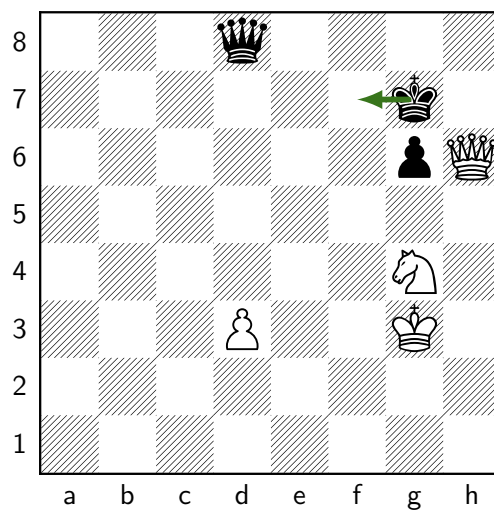
(d)

Figure 50: **Move 56, White: Kxd4.** (a) Our position; Melanie plays Kxd4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2553` (ply 120) [structural distance  $d=39$ , QAP obj=73]: the master played Kd6; under  $\mu$  our Kd3 plays the role of that game's Kd5; cross-type roles: our pd4 as their re2, our pa5 as their Rb3; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1465` (ply 106) [structural distance  $d=39$ , QAP obj=73]: the master played Kxd5; under  $\mu$  our Kd3 plays the role of that game's Kd4; cross-type roles: our rf5 as their Pa2, our qb3 as their nf5; transferred move d3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3400` (ply 95) [structural distance  $d=39$ , QAP obj=68]: the master played Kg7; under  $\mu$  our Kd3 plays the role of that game's Kh3; cross-type roles: our nc4 as their Pb6, our Nb1 as their pa4; transferred move d3d4 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kxd4 (summed QAP objective 508, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

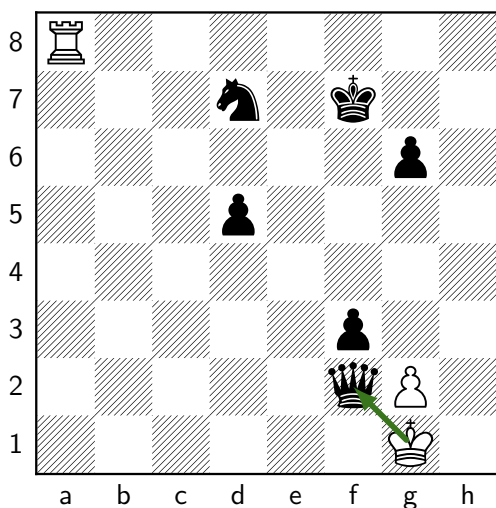




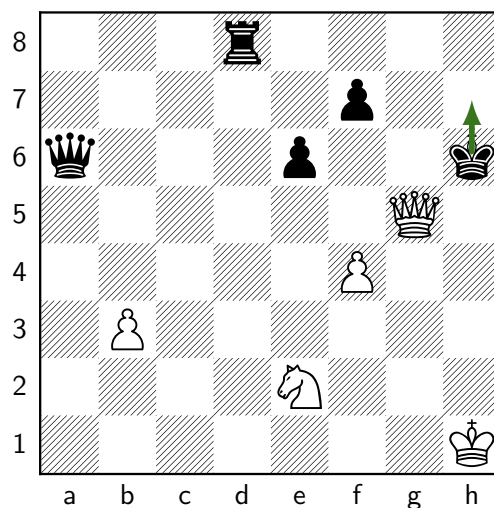
(a)



(b)



(c)



(d)

Figure 51: **Move 57, White: Kxc4.** (a) Our position; Melanie plays Kxc4 (d4c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#171` (ply 127) [structural distance  $d=39$ , QAP obj=75]: the master played Kf7; under  $\mu$  our *Kd4* plays the role of that game’s *Kg2*; cross-type roles: our *rf5* as their *Qd8*, our *Nb1* as their *Pg6*; transferred move d4c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1451` (ply 116) [structural distance  $d=39$ , QAP obj=74]: the master played Kxf2; under  $\mu$  our *Kd4* plays the role of that game’s *Kg1*; cross-type roles: our *Nb1* as their *Pg2*, our *nc4* as their *pf3*; transferred move d4c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#924` (ply 121) [structural distance  $d=37$ , QAP obj=74]: the master played Kh7; under  $\mu$  our *Kd4* plays the role of that game’s *Kh3*; cross-type roles: our *pa5* as their *Rd8*, our *rf5* as their *Pf7*; transferred move d4c4 (**endorses this move**). Synthesis: 43 retrieved analogs transferred to Kxc4 (summed QAP objective 1666, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.